

A parameter sweep of the moist V2 shallow-water Hadley model

149 simulations across the sign change of the gross moist stability, with off-equatorial and seasonally migrating forcing

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1 Summary of what the runs show

151 runs, of which 150 completed: 131 with time-invariant forcing and 19 with a seasonally migrating forcing, all at $n_y = 801$ except the resolution twins, all integrated 5800 days (5.01 drag timescales) except the seasonal ones and one short pair. One configuration diverged and was replaced.

1. **One number organises most of it.** $r = W_q/W^*$, the ratio of the moisture an undisturbed column holds to the moisture at which the gross moist stability changes sign, is the control parameter (Section 2). Four runs with a from 0.75 to 0.87 and W_c from 34.9 to 52.0, chosen so that all four sit at $r = 0.85$, agree on the subtropical jet to 0.19%, on peak $|v|$ to 0.10%, on the raining fraction to 0.00% and on peak precipitation to 1.02%. Across the 58 runs that vary the gross moist stability by any route, those same quantities span 77%, 382%, 59% and 839% of their means. The collapse is real, not a restatement of the definition.
2. **Two of the six parameters do not collapse, for identifiable reasons.** τ_c and E_0 each move r and also move something else: how fast convection removes water, and how much water there is. The τ_c ladder crosses $r = 1$ (0.906 to 1.084) with the domain raining everywhere at every rung, while a run at $r = 1.055$ reached through W_c rains over only 73% of the domain.

3. **The transition begins before $\hat{H}$ changes sign.** On the finest transect ($a = 0.85$, nine values of W_c) the domain rains everywhere up to $r = 0.919$ and has split into three bands over 91% of the domain by $r = 0.964$. Convergence raises W above W_q in the ascending region, so the local $\hat{H}$ goes negative there before the undisturbed column does.
4. **What the aggregated state does with its energy.** At $a = 0.85$, $W_c = 50$ the mean circulation carries $+104.7 \text{ MW m}^{-1}$ into the ascending column, because $\hat{H} < 0$ there, and the eddy moisture flux carries -106.2 back out. They cancel to 1.4%. The eddy flux is the brake, and the solution finds its equilibrium by drawing part of the domain dry enough to turn $\hat{H}$ positive: over $1.0 < |y| < 3.8 \text{ Mm}$ the column sits at 43.7 kg m^{-2} against $W^* = 44.25$, giving $\hat{H} = +1.0 \text{ MJ m}^{-2}$ where the rest of the domain has -33 (Section 6).
5. **Removing the eddy moisture flux removes the brake, completely.** At $D = 0$ on the negative-stability reference, 4.6% of the domain rains, at $831.7 \text{ mm day}^{-1}$, in 35 separate bands. At $D = 4 \times 10^6 \text{ m}^2 \text{ s}^{-1}$, 71.8% rains at a peak of 34.4. On the positive-stability reference the same range of D changes peak precipitation by 3%.
6. **Aggregation needs the feedback at nearly full strength.** With Λ at 0, 0.25, 0.50 and 0.75 of its physical value L_v/C the negative-stability reference rains everywhere, peaking at 4.9, 5.2, 5.7 and 7.9 mm day^{-1} . At the physical value it rains over 63% of the domain, peaking at 77.8. The onset between 0.75 and 1.0 is abrupt.
7. **With passive moisture the same parameter does nothing at all.** Two V1 twins, one at $r = 0.76$ and one at $r = 1.14$, have identical circulations: jet 28.20 m s^{-1} , peak $|v|$ 0.365 m s^{-1} , raining everywhere, and no organised band. Their gross moist stabilities are $+18.4$ and -11.4 MJ m^{-2} . $\hat{H}$ is a diagnostic in V1 and a dynamical quantity in V2, and this is the control that shows it.
8. **There is a second boundary, further out, where the solutions stop settling.** 18 of the 131 never reach a steady state. In 16 the jet magnitude itself swings, by 2.6 to 83.0 m s^{-1} over the trailing 2000 days, along with the peak rain rate by up to 2890 mm day^{-1} . In the other two the jet holds to under 0.7 m s^{-1} while the rain bands wander across the domain, which shows up as a pointwise u difference of 10% and 62% between consecutive 1000-day means. They span $r = 1.132$ to 2.290 , and the largest r that still settles is 1.310, so unsteadiness sets in well past $\hat{H} = 0$ rather than at it (Section 11).
9. **The initial moisture does not select the attractor.** Starting W 15 or 30 kg m^{-2} above W_c at four points spanning the transition changes the equilibrium by at most 8×10^{-4} relative. No multiple equilibria were found (Section 8).
10. **The rain band and the energy flux equator are 1.9 Mm apart.** With the forcing peak at 1400 km, the rain peaks at 1.082 Mm and the total column MSE flux vanishes at 2.959 Mm. Because $\hat{H}$ varies only from 17.8 to 19.3 MJ m^{-2} across the tropics there, the flux is essentially $v\hat{H}$ and vanishes where v does, which is the boundary between the cross-equatorial cell and the summer cell rather than the ascent maximum (Section 9).
11. **The seasonal response is quasi-static, and a single-timescale picture of it is wrong by a factor of ten.** A 37-day thermal relaxation predicts the rain band should lag the forcing by 17 to 37 days depending on period, and be damped to 0.36 of the quasi-steady amplitude at a 90-day period. Measured across five periods from 90 to 1440 days, the lag is 3.1 to 6.0 days and the gain 0.85 to 0.94. The gain also matches the steady off-equatorial

value at the same displacement to within 2%. These cycles repeat exactly: consecutive cycles differ by 0.0 km in band position (Section 10).

12. **Every seasonal run on the negative-stability reference is chaotic.** Consecutive cycles differ by 10,889 to 14,656 km in rain-band position, so their amplitudes and lags describe one arbitrary realisation and are not reported as results.
13. **None of it looks much like the observed Hadley circulation, and the jet is the worst of it.** Across the plane the subtropical jet runs 1.53 to 2.56 times the observed 31.0 m s^{-1} , and with the forcing 2800 km off the equator the winter jet reaches 128.7 m s^{-1} with -54 m s^{-1} of equatorial easterly. The eddy energy transport is essentially absent, as the ERA5 calibration already found (Section 5).
14. **The numerics are sound where it matters and not everywhere.** Halving Δt changes u by 0.01% of its peak. Across all eight resolution and timestep twins the jet moves by at most 0.30% and peak precipitation by at most 0.91%. But the profile at $|y| \approx 7$ to 11 Mm is not converged: between $n_y = 801$ and 401 it differs by up to 67.6% of peak u , and between 801 and 1601 by up to 9.8%. The tropics are converged; the westerly terminus region is not.

Independent check. Two of these configurations were run in July 2026 by a different session with different analysis code, and their equilibria were recorded in `CLAUDE.md` and in `docs/era5_calibration_report.pdf`. All 18 quantities recorded there are reproduced, most to under 1%: peak precipitation 77.82 against 78, the secondary band at 5.592 Mm and 18.72 mm day $^{-1}$ against 5.6 and 18.7, the dry fraction 0.362 against 0.36, the terminus notch at 7.482 Mm and -45.49 m s^{-1} against 7.48 and -45 , and the jet 73.79 against 73.8. The quiescent column sits 71.22 K above its radiative target against the analytic $\tau \Lambda E_0 = 71.22 \text{ K}$. Run `python scripts/v2_sweep_verify.py` to regenerate that comparison.

2 The gross moist stability, and the one number that sets its sign

2.1 What the quantity is

The moist V2 model carries column moist static energy $\langle h \rangle = C\theta + L_v W$, where C is the column heat capacity that converts a potential-temperature tendency into a column-energy tendency and L_v is the latent heat of vaporisation. Adding the thermodynamic equation (7) times C to the moisture equation (8) times L_v gives the budget (9), whose transport term is

$$\partial_y \left[\underbrace{v(\hat{S} - L_v(2a-1)W)}_{\text{mean circulation}} - \underbrace{L_v D \partial_y W}_{\text{eddies}} \right]. \quad (1)$$

The bracket multiplying v is the gross moist stability $\hat{H} = \hat{S} - L_v(2a-1)W$. It is the column energy the mean circulation carries per unit of upper-level meridional velocity, and it is the difference of two terms that partly cancel. The first, the gross dry stability $\hat{S} = C d \Delta_z / H$, is positive: the upper branch is warmer than the lower branch, so poleward flow aloft exports dry static energy. The second is negative: the lower branch is moister than the upper branch and flows the other way, so the same circulation imports latent energy.

Its *sign* is what matters, and here is why. Suppose ascent at some latitude strengthens. Ascent means upper-level divergence, $\partial_y v > 0$. If $\hat{H} > 0$, the strengthened circulation exports more energy from that column than it imports, the column cools, and the ascent is damped. If $\hat{H} < 0$ the

strengthened circulation is a net *import* of energy into the ascending column, which warms and moistens it further, and the ascent grows. A negative gross moist stability is a positive feedback on ascent. In a full model this is the mechanism behind self-aggregation of convection; here it is available in a two-line budget.

2.2 The two moisture values that decide the sign

$\hat{H}$ vanishes at

$$W^* = \frac{\hat{S}}{L_v(2a-1)} = \frac{C d \Delta_z}{H L_v(2a-1)}. \quad (2)$$

At the sweep's fixed values ($C = 5.1619 \times 10^6 \text{ J m}^{-2} \text{ K}^{-1}$, $d = 4 \text{ km}$, $\Delta_z = 60 \text{ K}$, $H = 16 \text{ km}$, $L_v = 2.5 \times 10^6 \text{ J kg}^{-1}$) this gives $\hat{S} = 7.7429 \times 10^7 \text{ J m}^{-2}$ and $W^* = 44.25 \text{ kg m}^{-2}$ at the spec's $a = 0.85$, rising to 53.40 at $a = 0.79$ and 61.94 at $a = 0.75$.

The other value is where the moisture actually sits. In a column with no transport, the steady moisture balance is $E_0 = P = (W - W_c)/\tau_c$, so

$$W_q = W_c + \tau_c E_0. \quad (3)$$

At the defaults $W_c = 50 \text{ kg m}^{-2}$, $\tau_c = 14400 \text{ s}$ and $E_0 = 4.6 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ the correction is $\tau_c E_0 = 0.66 \text{ kg m}^{-2}$, so $W_q = 50.66 \text{ kg m}^{-2}$. This value is set locally, by evaporation against precipitation, and the circulation cannot move it except by transporting water in or out faster than convection removes it. That is why the quiescent parts of the domain sit at W_q whatever the rest of the solution is doing.

2.3 One dimensionless number, five knobs

Putting (2) and (3) together, an undisturbed column has negative gross moist stability when $W_q > W^*$, that is when

$$r \equiv \frac{W_q}{W^*} = \frac{L_v(2a-1)(W_c + \tau_c E_0)}{C d \Delta_z / H} > 1. \quad (4)$$

Five separate model parameters appear in r , and each of them can be moved across $r = 1$ on its own: the lower-branch water-vapour fraction a , the precipitation threshold W_c , the convective timescale τ_c , the evaporation rate E_0 , and the static stability Δ_z . That is what makes the collapse onto r a real question rather than a definition. If r were merely a convenient relabelling, then runs at the same r reached through different parameters would still differ. The sweep therefore includes eight runs in which W_c was chosen for each of four values of a so that all four land on the same r , at $r = 0.85$ and again at $r = 1.05$ (Section 3), and separate ladders in τ_c , E_0 and Δ_z that cross $r = 1$ by themselves.

Two caveats on r , stated now because they limit how much it can be expected to explain. First, r describes an *undisturbed* column. In the deep tropics the circulation moves water, so the local W departs from W_q and the local $\hat{H}$ from its quiescent value; r predicts the sign of the feedback that gets things started, not the equilibrium reached. Second, a and Δ_z appear elsewhere in the equations as well as in r . The factor $(2a-1)$ is also the moisture transport velocity in (8), and Δ_z also multiplies the adiabatic term in (7), so changing either does more than move r . W_c , τ_c and E_0 are cleaner: they enter the dynamics only through r and through the precipitation rate itself.

2.4 Where the real atmosphere sits

This is not an exotic corner of parameter space. The ERA5 calibration ([docs/era5_calibration_report.pdf](#), its stability section) gives $\hat{H}/\hat{S}$ between 0.258 over $|\varphi| \leq 10^\circ$ and 0.318 over $|\varphi| \leq 30^\circ$, so the observed band-mean gross moist stability is positive but only about a quarter to a third of the dry value: the two competing terms cancel to within 70% or so. The same report finds the *local* $\hat{H}$ falling through zero within about 8° of the equator. And its estimate of W^* itself, 44 to 57 kg m^{-2} depending on how the time mean is taken, straddles the model's quiescent 50.66 kg m^{-2} . So $r = 1$ is roughly where the observations put the tropical atmosphere, and the question of what happens on either side of it is a question about the real Hadley circulation.

3 Where the regime boundary sits

3.1 The map

Figure 1 places 25 runs in the plane of a and W_c , the two parameters that move r without touching anything else in the equations. The line $r = 1$ is drawn on it, and it separates the two behaviours cleanly. Below it the whole domain rains, at close to the evaporation rate, with a single broad maximum. Above it the rain concentrates into a few narrow bands separated by regions where precipitation is identically zero, and the peak rate rises by more than an order of magnitude.

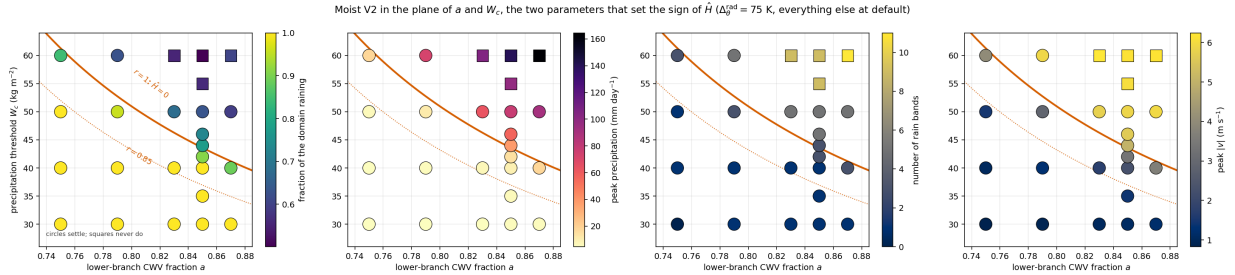


Figure 1: The regime map. Colour shows four properties of the equilibrium; the solid line is $r = 1$, where an undisturbed column has $\hat{H} = 0$, and the dotted line is $r = 0.85$. Circles settle to a steady state, squares never do.

3.2 The transect through the boundary

The finest sampling is at $a = 0.85$, where nine values of W_c from 30 to 60 put r between 0.693 and 1.371:

W_c	r	raining fraction	bands	peak P (mm day^{-1})	min $\hat{H}$ (MJ m^{-2})	jet (m s^{-1})
30	0.693	1.000	1	5.30	+23.4	51.5
35	0.806	1.000	1	6.15	+14.4	55.1
40	0.919	1.000	1	9.02	+4.8	60.8
42	0.964	0.910	3	15.02	-0.5	64.7
44	1.009	0.775	3	38.04	-10.7	70.5
46	1.055	0.730	5	56.64	-19.6	74.1
50	1.145	0.633	5	77.82	-32.8	73.8
55	1.258	0.561	9	97.67	-47.3	74.5
60	1.371	0.501	9	139.52	-68.0	78.3

The boundary is bracketed between $r = 0.919$ and $r = 0.964$, so it sits *below* $\hat{H} = 0$ for the undisturbed column. The reason is in the caveat of Section 2: r describes a column with no transport, and the ascending region has transport. Moisture convergence lifts W there above W_q , so the local $\hat{H}$ reaches zero while the quiescent collar still has $\hat{H} > 0$. At $W_c = 42$, $r = 0.964$, the minimum $\hat{H}$ over the domain is already -0.5 MJ m^{-2} .

3.3 The collapse, and its two exceptions

Figure 2 plots nine properties of the equilibrium against r for every run that varies the gross moist stability by any route. Four routes collapse: a , W_c , Δ_z , and the matched- r block that fixes r and varies a with W_c compensating.

The matched- r block is the test that could have failed. Eight runs, four at $r = 0.85$ and four at $r = 1.05$, with a from 0.75 to 0.87 and W_c chosen for each to hit the target r :

	raining fraction	peak P	jet	peak $ v $	min $\hat{H}$
$r = 0.85$, spread across four a	0.00%	1.02%	0.19%	0.10%	2.71%
$r = 1.05$, spread across four a	0.34%	14.91%	2.27%	0.07%	41.65%
for scale: spread over all 58 GMS runs	58.6%	838.6%	77.5%	382.3%	1877.9%

Each spread is the full range divided by the mean. At $r = 0.85$ the four runs are the same solution to a fraction of a percent, while a differs by 16% and W_c by 49% between them. At $r = 1.05$ the agreement is looser and structured: the circulation still matches (peak $|v|$ to 0.07%, jet to 2.3%) while the precipitation extremes do not (peak P to 15%, min $\hat{H}$ to 42%). Past the boundary the solution has narrow bands whose peak values are sensitive to where exactly they sit, and r predicts the circulation better than it predicts the extremes.

The two exceptions are informative rather than a failure of the idea. τ_c enters r through $W_q = W_c + \tau_c E_0$, and it also sets how fast convection removes water from a column. Raising it from 1800 to 172800 s at $a = 0.85$, $W_c = 40$ takes r from 0.906 to 1.084, across the boundary, and the domain rains everywhere at every rung, peak precipitation only rising from 8.1 to 17.0 mm day⁻¹. A slow convective sink smears the rain out, which is the opposite of what concentrating it requires, so the two effects work against each other. E_0 likewise enters r only weakly, through the same product, while setting the total water supply outright: doubling it on the negative-stability reference takes peak $|v|$ from 5.86 to 11.15 ms⁻¹ and peak precipitation from 77.8 to 197.5 mm day⁻¹ at nearly unchanged r (1.145 to 1.160).

Δ_z collapses for the energetics and the raining fraction, and sits about 6% low in the jet at matched r , because it also multiplies the adiabatic term in the thermodynamic equation. That was the expected behaviour of the two parameters flagged in Section 2 as doing more than moving r .

4 What the solutions on either side look like

4.1 Below the boundary: a Hadley cell that drizzles

At $r < 0.92$ the solution is a recognisable two-cell Hadley circulation with a broad rain maximum at the equator sitting on a background that rains everywhere at the evaporation rate. W is nearly flat, pinned close to W_q , with a bump of a few kg m⁻² at the equator. Peak precipitation is 5 to 9 mm day⁻¹ against the 3.97 the uniform evaporation supplies, so the equatorial maximum is only a 30 to 130% enhancement over the background. The jet strengthens smoothly with r , from 51.5 to 60.8 ms⁻¹ over the transect, and peak $|v|$ from 1.10 to 2.27 ms⁻¹.

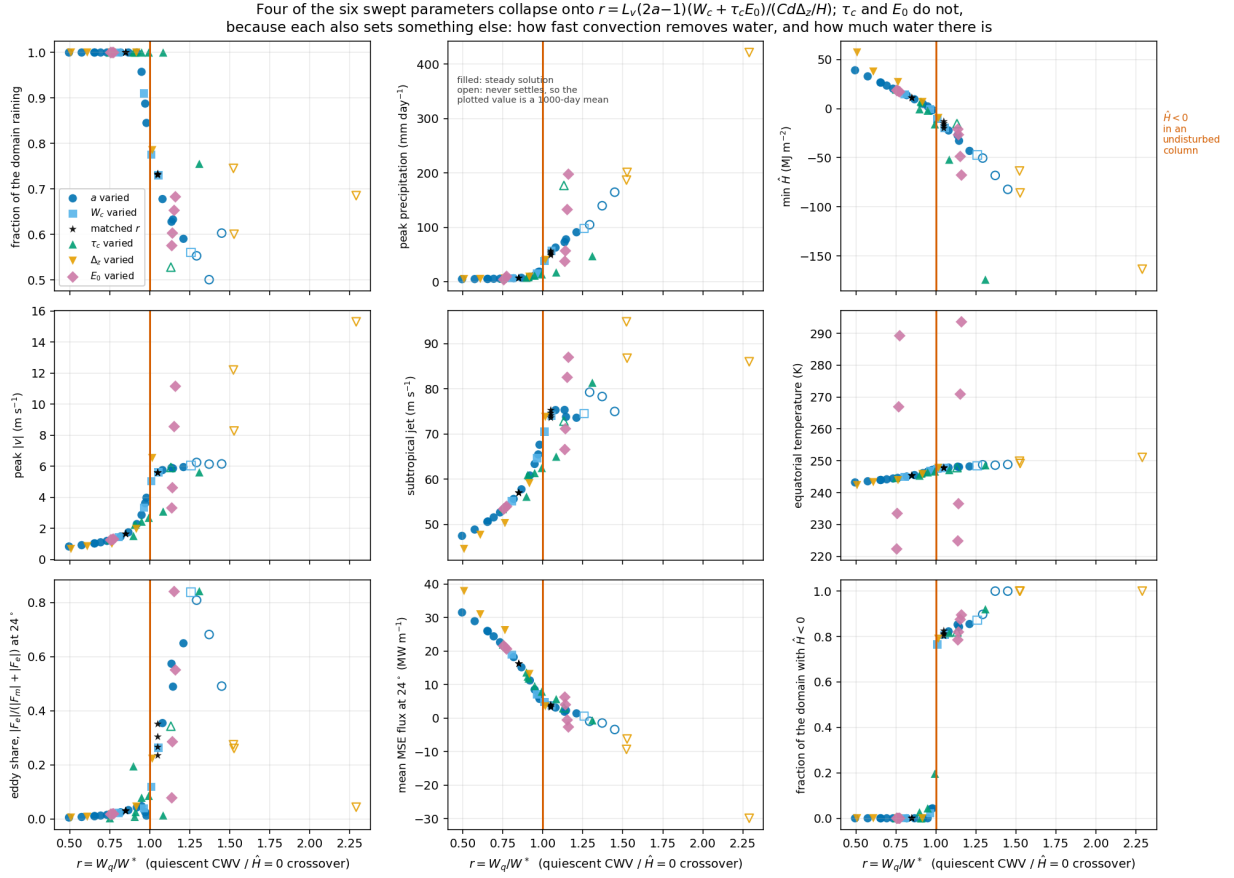


Figure 2: Nine properties of the equilibrium against r , for every run that varies the gross moist stability. Marker shape and colour identify which parameter was varied. Filled markers settle to a steady state; open markers never do, so their plotted value is a 1000-day time mean of a fluctuating state.

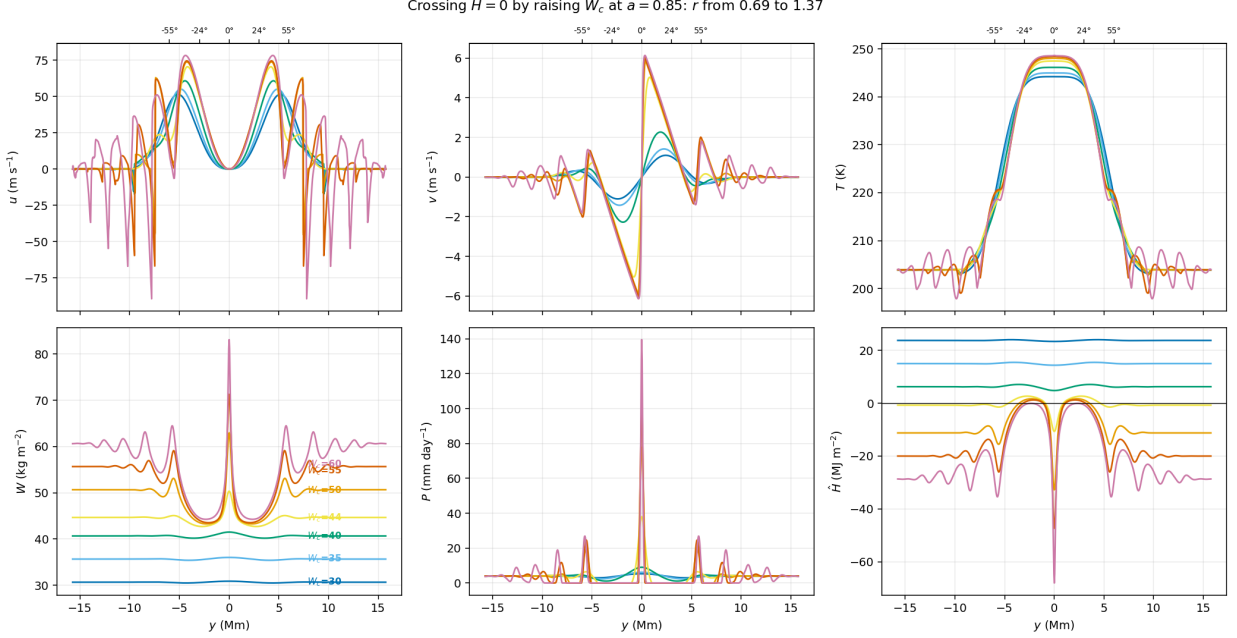


Figure 3: The $a = 0.85$ transect at equilibrium, W_c from 30 to 60, which is r from 0.693 to 1.371. Labels on the W panel sit in the quiescent collar, where $W = W_c + \tau_c E_0$ and the curves are furthest apart. The two largest W_c never settle, and their large extratropical oscillations in u are discussed in Section 11.

4.2 Above it: bands, dry gaps, and rain rates no atmosphere has

At $r > 1$ the picture is different in kind. At $W_c = 50$ the equator rains at 77.8 mm day^{-1} in a spike a few hundred km wide, two secondary bands sit at $\pm 5.59 \text{ Mm}$ raining 18.7 mm day^{-1} , and precipitation is identically zero over 36.2% of the domain. As r rises further the number of bands goes 3, 5, 9 and the peak rate reaches $139.5 \text{ mm day}^{-1}$ at $r = 1.371$ and 831.7 with the eddy moisture flux switched off entirely.

Two arithmetic facts make this less mysterious than it first looks, and both are exact. Peak precipitation is $(W_{\max} - W_c)/\tau_c$, so it is a direct read-out of the wettest column and nothing more. And because transport moves no total water, the domain mean of P equals E_0 at equilibrium, which the runs satisfy to between 0.999 and 1.000 in the finite-volume measure. Aggregation therefore cannot change how much it rains in total; it can only change over how little of the domain that rain falls. At $W_c = 50$ the raining 63.3% of the domain averages 1.58 times E_0 , and the excess is concentrated in the spike.

The precipitation rates are far outside anything observed. A zonal-mean 77.8 mm day^{-1} is roughly ten times the wettest zonal-mean band on Earth, and 831.7 is not a rain rate at all. This is the clearest sense in which the negative-stability solutions are not candidate Hadley circulations.

5 The solutions against ERA5

Before any of the parameter dependences matter, the question is whether these are plausible Hadley circulations. The short answer is that the positive stability side is the right order of magnitude and too strong by roughly a factor of two, the negative stability side is not a Hadley circulation, and the eddy energy transport is missing from all of it.

a	W_c	r	jet (ms^{-1})	peak $ v $ (ms^{-1})	$\Delta\theta(16^\circ)$ (K)	(24°) (K)	(33°) (K)
ERA5			31.0	1.07–1.31	0.56	3.21	8.75
0.75	30	0.49	47.5	0.83	0.38	2.34	8.31
0.79	40	0.76	53.6	1.27	0.56	3.33	11.33
0.85	40	0.92	60.8	2.27	0.92	5.07	15.96
0.85	44	1.01	70.5	5.03	1.60	7.33	20.91
0.85	50	1.14	73.8	5.86	1.71	7.62	21.62
V1 twin, $a=0.79$ $W_c=40$			28.2	0.37	0.18	1.13	4.16

The ERA5 row is the 2000–2009 annual mean from `docs/era5_calibration_report.pdf` (its Table 9), with peak $|v|$ quoted as a slab velocity at the two branch depths that report estimates, because that is the quantity the model’s v is. Temperature contrasts are $\theta(0) - \theta(y)$ at the Coriolis-mapped latitudes $y = 2, 3$ and 4 Mm.

The jet is too strong everywhere, by 1.53 to 2.56 times. Over the whole (a, W_c) plane the subtropical jet runs between 47.5 and 79.2 ms^{-1} against an observed 31.0. The best case in the sweep is the driest, $a = 0.75$, $W_c = 30$ at $r = 0.495$, which gives 47.5 ms^{-1} . The ERA5 calibration already traced this overshoot to the radiative retarget rather than to the moisture parameters, and the attribution chain here confirms it quantitatively: the V1 twin has a 28.20 ms^{-1} jet, retargeting the relaxation to $\Delta_\theta^{\text{rad}} = 75$ K with the feedback severed raises it to 41.58 with no change in moisture physics at all, and the sensitivity is 0.708 ms^{-1} per K of $\Delta_\theta^{\text{rad}}$ on the positive-stability reference. Getting the jet to 31 ms^{-1} at fixed everything-else would take $\Delta_\theta^{\text{rad}} \approx 43$ K, which is below the radiative-convective 50 K and so is not available.

The temperature contrast is too steep by 1.5 to 2.4 times, in the same direction and for the same reason. At 24° the observed $\Delta\theta$ is 3.21 K; the runs give 2.34 at $r = 0.49$, 3.33 at 0.76, 5.07 at 0.92 and 7.62 at 1.14. The one run that matches the observed contrast closely is the V1 twin at 1.13 K, which is too flat.

Column water vapour is about right on the dry side. ERA5 gives 39.6 kg m^{-2} averaged over $|\varphi| \leq 20^\circ$. The runs give 40.8 at $r = 0.76$ and 41.0 at $r = 0.92$, so the reference states hold close to the observed tropical column. That is unsurprising, since W is pinned near W_q and W_q was chosen near observed values, but it does mean the moisture is not the thing that is wrong.

The energy transport is the largest gap, and it is structural. The column fluxes at 24° , which is $y = 3$ Mm:

a	W_c	r	mean $v\hat{H}$ (MW m^{-1})	eddy $-L_v D\partial_y W$ (MW m^{-1})	total (MW m^{-1})
ERA5			27.9	57.7	85.6
0.75	30	0.49	31.54	0.16	31.70
0.79	40	0.76	21.25	0.38	21.63
0.85	40	0.92	11.24	0.50	11.74
0.85	44	1.01	4.86	-0.65	4.21
0.85	50	1.14	2.29	-2.19	0.10
V1 twin, $a=0.79$ $W_c=40$			6.73	0.09	6.82

At 24° ERA5 transports 85.6 MW m^{-1} of column moist static energy northward, of which 57.7 is eddy and 27.9 is the mean circulation. The positive-stability reference gives 21.3 mean and 0.38 eddy. So the mean flux is within a factor of one and a half of the observed mean flux, and the eddy flux is smaller than observed by a factor of 150. The reason is the one the ERA5 report identified: the model's only eddy energy flux is $-L_v D \partial_y W$, there is no eddy dry-static-energy flux in the equations at all, and W is nearly flat because the local evaporation-precipitation balance pins it, so there is almost no gradient for D to act on. Raising D to $4 \text{ million m}^2 \text{ s}^{-1}$, above the top of the ERA5-calibrated range, moves the eddy flux from 0.38 to 1.42 MW m^{-1} .

The negative-stability side inverts the problem rather than fixing it: at $a = 0.85$, $W_c = 50$ the eddy flux at 24° is -2.19 MW m^{-1} , pointing the wrong way, because W decreases away from the ITCZ there and the downgradient flux is equatorward.

6 The column energy budget

6.1 Never the net alone

The mean column MSE flux $v \hat{H}$ is the difference of two much larger terms, and reporting it alone would misrepresent the transport by a large factor. At 24° on the positive-stability reference the dry-static flux $\hat{S}v$ and the latent flux $-L_v(2a - 1)vW$ nearly cancel, and the ERA5 calibration measured the same cancellation in the observations (its Table 8: $+50.7$ against -22.9 observed, $+114.5$ against -100.5 in the model at $a = 0.77$). The model over-transportes each component and under-transportes their difference.

6.2 How a negative-stability solution closes its budget

Figure 4 answers this for $a = 0.85$, $W_c = 50$, and the answer has two parts.

The eddy flux cancels the mean flux at the ITCZ, to 1.4%. At the ascending column the mean circulation carries $+104.7 \text{ MW m}^{-1}$ northward on one side of the spike and the same southward on the other, importing energy because $\hat{H} < 0$ there. The eddy moisture flux carries -106.2 , exporting it down the enormous W gradient the spike creates. The total is -1.5 MW m^{-1} , 1.4% of either term. This is what stops the positive feedback: the ascent intensifies until the moisture gradient it builds is steep enough for D to remove the energy the mean circulation is importing. That is also why D is the parameter with the most leverage on this state (Section 7).

The circulation dries part of the domain back above $\hat{H} = 0$. Over $1.0 < |y| < 3.8 \text{ Mm}$ the column is drawn down to a mean 43.7 kg m^{-2} , below $W^* = 44.25$, so $\hat{H}$ there is $+1.0 \text{ MJ m}^{-2}$ against -32.8 at the ITCZ, reaching $+1.70$ at 2.28 Mm . Those bands are the only part of the domain from which the mean circulation can export energy, and they lie inside the non-raining region.

That second part is weaker than the picture suggests at a glance, and the number is worth stating rather than the impression: 43% of the non-raining area has $\hat{H} > 0$, and averaged over the whole non-raining area $\hat{H}$ is -2.54 MJ m^{-2} . The two broad gaps between the ITCZ and the secondary bands are positive; the narrow gaps outside the secondary bands are not. So the drying is part of the answer and the eddy cancellation is most of it.

6.3 Budget closure as an equilibrium check

For the 113 runs that settle, the residual of $\partial_y(F_{\text{mean}} + F_{\text{eddy}}) = Q$ is between 3.4×10^{-9} and 6.3×10^{-4} of the maximum source, and 110 of them are below 10^{-6} . The three above it are

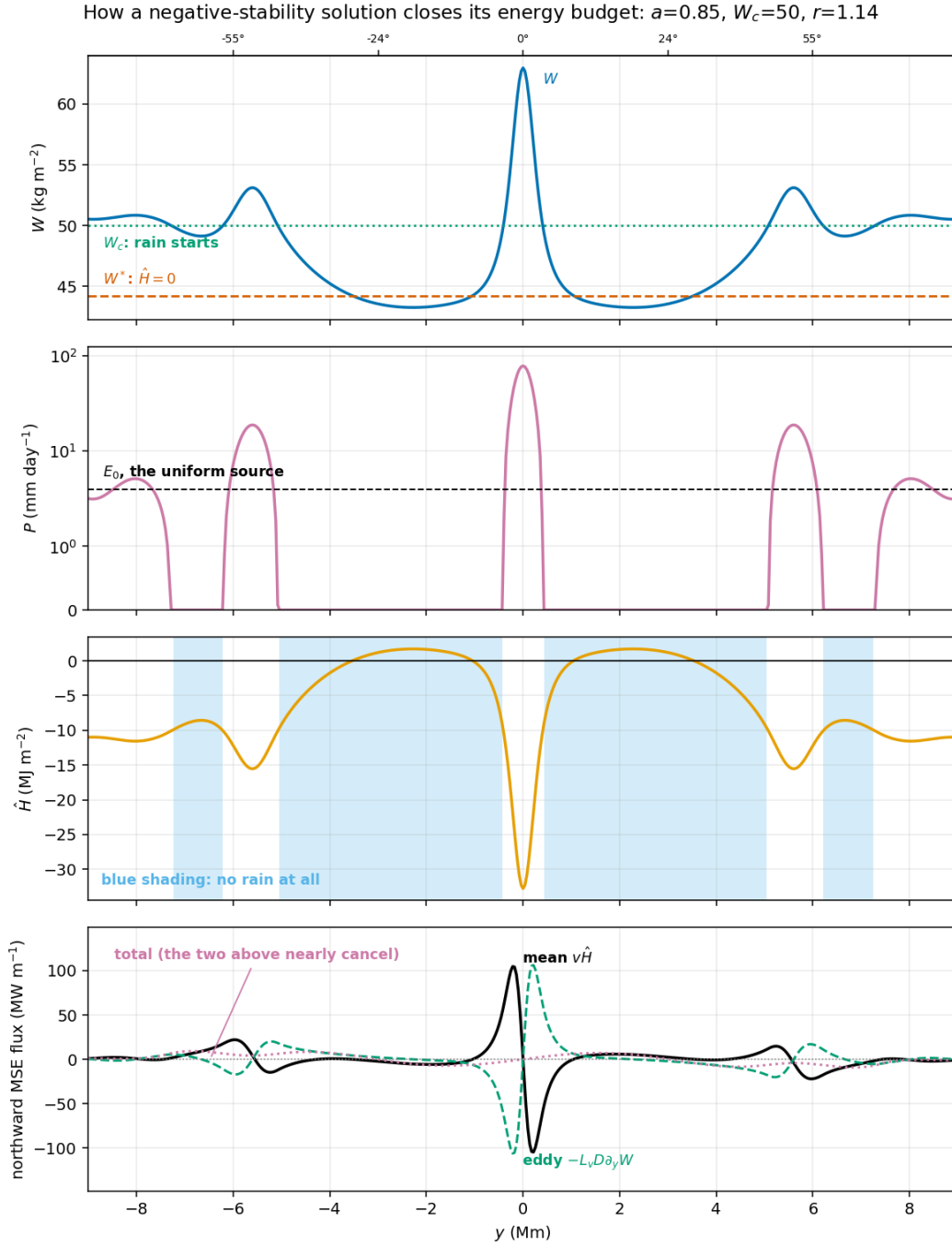


Figure 4: How the negative-stability solution at $a = 0.85$, $W_c = 50$ ($r = 1.145$) reaches equilibrium. Top to bottom: column water vapour against the two thresholds that matter, precipitation on a log scale, the gross moist stability with the non-raining regions shaded, and the column MSE flux resolved into its mean and eddy parts.

identifiable: the two members of the off-equatorial resolution pair, which were run 2500 days rather than 5800 and so are not fully equilibrated (5×10^{-6} and 2×10^{-6}), and the $\tau_c = 1800$ s run at 6.3×10^{-4} , whose u drift is 2×10^{-5} so it is settled; its residual is presumably the diagnostic reconstruction struggling with a precipitation that is a very stiff function of W , and I did not chase it further.

For the 18 runs that never settle the residual runs from 0.12 to 1690, which is expected rather than a defect: $\partial\langle h \rangle / \partial t$ is not zero and the budget as written omits it. Their residual is therefore a useful independent confirmation of the steadiness classification, not a separate check.

7 Sensitivity to each parameter in turn

Each ladder is run on both reference states, the positive-stability $a = 0.79$, $W_c = 40$ ($r = 0.76$) and the negative-stability $a = 0.85$, $W_c = 50$ ($r = 1.14$). The figures are 05_ladder_dyrad.png through 10_ladder_evap.png.

7.1 The radiative contrast $\Delta_\theta^{\text{rad}}$: the strongest single lever on the jet

Raising $\Delta_\theta^{\text{rad}}$ from 50 to 110 K takes the jet from 35.8 to 78.3 ms^{-1} on the positive-stability reference and from 57.6 to 99.3 on the negative one. The sensitivity is 0.708 and 0.690 ms^{-1} per K, nearly the same on both, so this is a dry-dynamical response that the moisture regime rides on top of rather than modifies. Peak $|v|$ goes 0.79 to 2.03 ms^{-1} on the positive reference. The raining fraction stays 1.000 throughout on the positive side and falls only from 0.648 to 0.591 on the negative side, so forcing strength does not by itself trigger aggregation over this range.

Physical reading (derived). $\Delta_\theta^{\text{rad}}$ sets the meridional temperature gradient the relaxation drives toward, the pressure gradient in (6) follows it, and the jet follows by thermal wind. That the sensitivity is the same on both sides of the stability boundary says the two effects are close to separable: latent heating changes where the rain falls and how strong the overturning is, and $\Delta_\theta^{\text{rad}}$ sets the overall amplitude.

7.2 The eddy moisture diffusivity D : the brake, and only on one side

This is the largest asymmetry in the sweep. On the positive-stability reference, D from 0 to $4 \times 10^6 \text{ m}^2 \text{ s}^{-1}$ changes the jet from 53.66 to 53.31 ms^{-1} (0.7%) and peak precipitation from 5.77 to 5.61 mm day^{-1} (3%). On the negative-stability reference the same range gives:

$D \text{ (m}^2 \text{ s}^{-1}\text{)}$	raining fraction	bands	peak $P \text{ (mm day}^{-1}\text{)}$	$W \text{ at equator (kg m}^{-2}\text{)}$
0	0.046	35	831.7	188.6
2.5×10^5	0.538	13	212.8	85.5
5×10^5	0.568	5	127.2	71.2
1×10^6	0.633	5	77.8	63.0
2×10^6	0.693	3	52.0	58.7
4×10^6	0.718	3	34.4	55.7

Physical reading (derived). Section 6 showed that the equilibrium at $D = 10^6$ is a near-cancellation between a mean flux importing +104.7 MW m^{-1} into the ascending column and an eddy flux exporting −106.2. The eddy flux is proportional to D , so at smaller D the ascent has to build a steeper moisture gradient to remove the same energy, which means a taller and narrower

spike. At $D = 0$ there is nothing to remove it at all, and the solution collapses to 4.6% of the domain raining at 832 mm day^{-1} with the equatorial column holding 189 kg m^{-2} of water. That run also fails the steadiness test through wandering bands rather than a wandering jet.

Two things follow. The parameter with the most leverage over the aggregated state is the one the ERA5 calibration is least certain about; its measured range is 3×10^5 to 4×10^6 , and over that range the raining fraction goes 0.54 to 0.72. And the positive-stability reference is almost completely insensitive to it, so D cannot be calibrated from anything on that side.

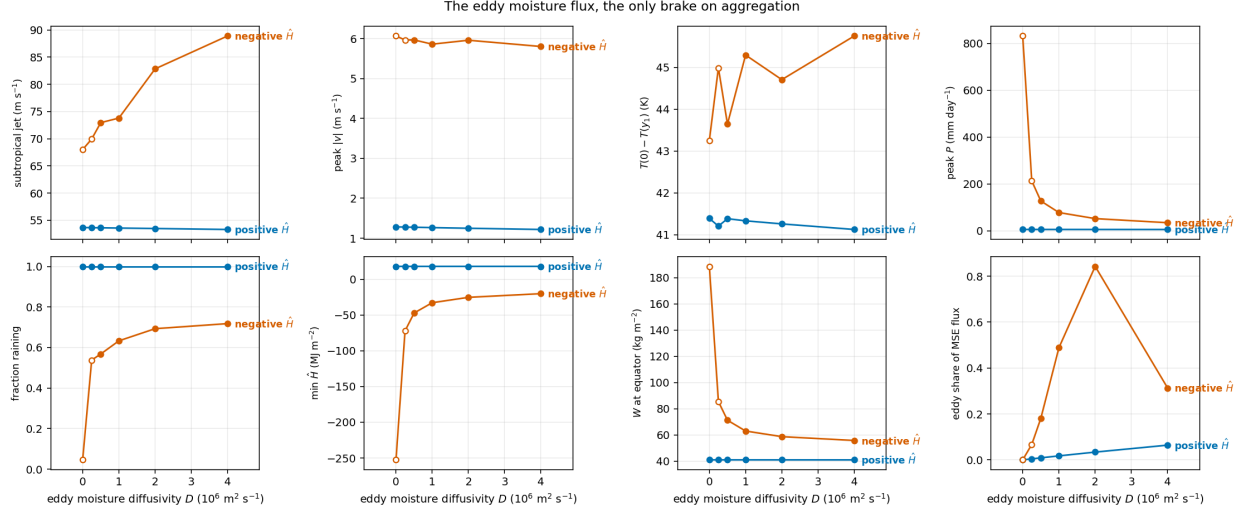


Figure 5: The eddy moisture diffusivity ladder on both reference states. Hollow markers are runs that never settle. The positive-stability curves are flat in every panel; the negative-stability ones are not.

7.3 The convective timescale τ_c : aggregation needs a fast sink

On the negative-stability reference, τ_c from 1800 to 172800 s takes the raining fraction from 0.528 to 0.755 and peak precipitation from 176.8 down to 46.9 mm day^{-1} , so a *shorter* convective timescale aggregates more, even though a shorter τ_c gives a *smaller* r . This is the mechanism behind the failed collapse of Section 3: concentrating rain requires the sink to respond quickly to the local moisture, and a slow sink smears it out. The $\tau_c = 1800 \text{ s}$ run is one of the 18 that never settles.

7.4 Evaporation E_0 : the amplitude of everything moist

E_0 barely moves r (1.138 to 1.160 over a fourfold range on the negative-stability reference) and moves everything else a great deal: peak $|v|$ from 3.31 to 11.15 m s^{-1} , peak precipitation from 37.4 to $197.5 \text{ mm day}^{-1}$, and the equatorial temperature from 224.9 to 293.6 K . The temperature is the cleanest demonstration of the mean-state offset derived in SCIENCE.md §3.6: the quiescent column sits $\tau \Lambda E_0$ above its radiative target, which is 35.6 K at the lowest E_0 and 142.4 at the highest, and over eight runs spanning that range the measured offset matches the prediction to 4×10^{-7} relative. On the positive-stability reference E_0 changes the jet by 1% while changing peak precipitation from 3.69 to 9.80 mm day^{-1} , exactly tracking E_0 itself because the domain rains uniformly at the evaporation rate.

7.5 The eddy momentum flux v_d : the axisymmetric limit is extreme

Removing the eddy momentum flux entirely gives a jet of 86.7 ms^{-1} on the positive-stability reference and 160.9 ms^{-1} on the negative one, against 53.6 and 73.8 at the default $v_d = 2.5 \text{ ms}^{-1}$. Peak $|v|$ collapses from 1.27 to 0.31 ms^{-1} on the positive side, and raising v_d to 7.5 gives 47.5 ms^{-1} of jet with 2.07 ms^{-1} of overturning. All four directions match the expectations in SCIENCE.md §6: stronger eddies give weaker zonal winds, stronger v , and a wider cell.

At $v_d = 0$ the negative-stability run still aggregates, with 76.5% of the domain raining and a peak of 72.7 mm day^{-1} . So aggregation is thermodynamic, and the eddy momentum forcing modulates it rather than causing it.

7.6 The feedback strength Λ : a threshold, not a gradual onset

$\Lambda = L_v/C$ is derived rather than free, and detuning it breaks the exact cancellation of precipitation in the MSE budget, so this ladder is diagnostic rather than physical. It is also the most informative one. On the negative-stability reference:

Λ/Λ_0	jet (ms^{-1})	peak $ v $ (ms^{-1})	peak P (mm day^{-1})	raining fraction
0.00	41.58	0.562	4.91	1.000
0.25	44.69	0.683	5.16	1.000
0.50	48.86	0.912	5.70	1.000
0.75	57.36	1.691	7.93	1.000
1.00	73.79	5.860	77.82	0.633
1.25	81.90	7.526	166.85	0.620

Between 0.75 and 1.00 the peak rain rate jumps by a factor of 9.8 and the raining fraction falls off 1. On the positive-stability reference the same ladder never aggregates: the raining fraction is 1.000 at every rung and peak precipitation goes 4.59 to $10.27 \text{ mm day}^{-1}$. So the two ingredients are both necessary: negative gross moist stability, and a feedback strong enough to exploit it. At $\Lambda/\Lambda_0 = 0$ the two references have identical dry fields to every digit reported (jet 41.5779 ms^{-1} , peak $|v|$ 0.562137 ms^{-1} , $T(0)$ 197.123 K) while their moisture differs as their W_c requires, which is the expected behaviour of the V20 bridge and a check that the coupling runs only in the intended direction.

7.7 The V1 to V2 attribution chain

Four runs per reference, changing one thing per step:

$\hat{H}$	step	jet (ms^{-1})	peak $ v $ (ms^{-1})	$T(0)$ (K)	$\Delta\theta_{24}$ (K)	peak P (mm day^{-1})
either	V1, no heating, no retarget	28.20	0.365	200.3	1.13	4.38
either	retarget to 75 K, $\Lambda = 0$	41.58	0.562	197.1	1.63	4.59
+	Λ on, target still 50 K	35.84	0.790	246.9	2.38	5.08
+	Λ on, target 75 K (V2)	53.57	1.266	244.6	3.33	5.72
−	Λ on, target still 50 K	57.56	5.242	249.9	7.05	67.47
−	Λ on, target 75 K (V2)	73.79	5.860	248.1	7.62	77.82

The first two rows have no moisture feedback, so they are the same run whichever W_c is set; the peak P differs only because P reads off W_c .

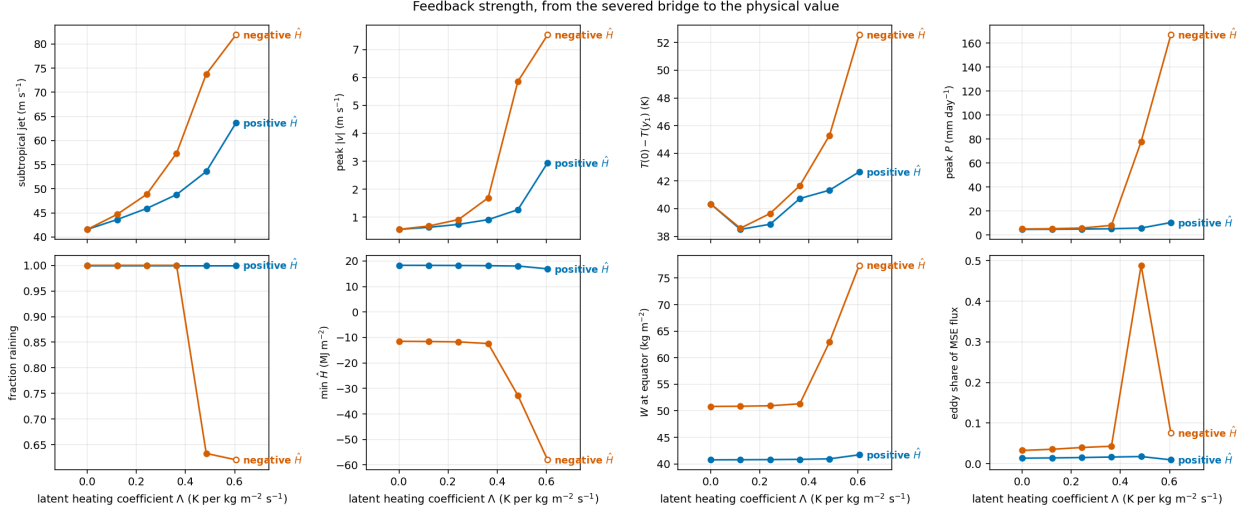


Figure 6: The latent-heating coefficient ladder. The physical value is $\Lambda_0 = L_v/C = 0.4843 \text{ K per kg m}^{-2} \text{ s}^{-1}$, the fifth point from the left. On the negative-stability reference everything jumps between the fourth and fifth points; on the positive-stability one nothing does.

The retarget alone accounts for a 47% jet increase with no moisture feedback at all, which is where most of the overshoot against observations comes from. The 71 K warming appears with Λ , as derived. And aggregation appears in the negative-stability row as soon as Λ is switched on, even at the flatter 50 K target, so it does not require the steeper radiative contrast.

8 Two equilibria at the same parameters?

A state whose ascent reinforces itself is a natural candidate for multiple equilibria, so eight runs start with the column already moist: at $a = 0.85$ and $W_c = 40, 42, 44$ and 46 , spanning r from 0.919 to 1.055 across the transition, with $W(t=0) = W_c + 15$ and $W_c + 30 \text{ kg m}^{-2}$ instead of the default W_c .

Nothing happens. The largest relative difference from the default start, over any of the reported quantities and any of the eight runs, is 8×10^{-4} . At $W_c = 40, 42$ and 44 the equilibria agree to every digit reported: jet $60.81, 64.68, 70.50 \text{ ms}^{-1}$ and peak P $9.02, 15.02, 38.04 \text{ mm day}^{-1}$ whatever the initial moisture. At $W_c = 46$ the jet differs by 0.06 ms^{-1} out of 74.07 and peak precipitation by 0.04 out of 56.64 .

So within the range tested there is one attractor per parameter set, and the aggregated state is reached from a dry start as readily as from a wet one. That is a negative result and worth having: it means the regime map of Section 3 is a map of the parameter space rather than of the initial conditions, and the boundary at $r \approx 0.94$ is not a hysteresis loop. What it does not test is a start that is drier than W_c , or an asymmetric initial perturbation, either of which could still find a second branch (Section 12).

9 Forcing peaked off the equator and held fixed in time

Sixteen runs move the maximum of the radiative target off the equator and hold it there, at $y_0 = 0, 700, 1400, 2100$ and 2800 km . Two profile families are used. The $\sin^2$ family translates rigidly,

so y_0 displaces the forcing without changing its shape or strength. The SB08 family adds an antisymmetric term, so its peak value *rises* above θ_{00} by $\Delta_\theta^{\text{rad}} \sin^2(\pi y_0/2y_1)$ as y_0 increases: a solstitial SB08 forcing is stronger as well as displaced, which is worth holding in mind when its circulation is stronger too. Both families are run on both reference states.

9.1 Where the rain goes, and how to ask the question

Three positions, and they are not interchangeable. The precipitation centroid taken over the whole domain is a bad tracker here, by enough to change the answer. On the positive-stability reference the domain drizzles at the evaporation rate everywhere, and that uniform background pulls the centroid toward the middle of the domain: at $y_0 = 1400$ km the plain centroid sits at 0.299 Mm while the rain *maximum* sits at 1.082 Mm. Read from the centroid the rain band follows 21% of the forcing displacement, read from the maximum 77%. The report uses the maximum, and a centroid taken over the connected region raining above E_0 , which agree with each other. All three are in `all_runs.csv`, so the difference is visible rather than buried.

The rain follows most but not all of the forcing. On the positive-stability reference the rain maximum reaches 1.082, 1.404 and 1.711 Mm for SB08 forcing at $y_0 = 1400, 2100$ and 2800 km, so it covers 77%, 67% and 61% of the displacement. The fraction falls as the forcing goes further out, which is what a restoring influence tied to the equator produces. Under $\sin^2$ forcing the fractions are slightly higher, 82% at 1400 km and 70% at 2100 km.

9.2 The rain maximum and the energy flux equator are 1.9 Mm apart

Take the SB08 $y_0 = 1400$ km run on the positive-stability reference. Its rain peaks at 1.082 Mm. The zero of its total column MSE flux, the energy flux equator, is at 2.959 Mm. The two are separated by 1.88 Mm, which is about 15° of equivalent latitude.

The reason is visible in the flux itself. The gross moist stability in that run varies only between 17.8 and 19.3 MJ m^{-2} across the whole tropics, and the eddy flux never exceeds 0.8 MW m^{-1} against a mean flux reaching 80 MW m^{-1} . So the total flux is very nearly $v\hat{H}$ with $\hat{H}$ constant, which means it vanishes where v does. And v has no zero anywhere between -5.0 and $+3.0$ Mm: a single cross-equatorial cell spans the entire tropics with southward upper-level flow throughout, and the rain maximum sits *inside* that cell rather than at its edge. Upper-level divergence $\partial_y v$ is positive across the whole band, peaking at 1.061 Mm, which is where the rain is (1.082 Mm, agreeing to 21 km). The flux zero at 2.959 Mm is the boundary between that cell and the weak summer cell poleward of it.

So in this model the standard reasoning that puts the rain band at the energy flux equator does not apply, and the reason is structural: the ascending region is broad and lies within a single-signed v , so the energy flux does not have to vanish where the ascent is strongest. Whether that is a defect of the single-layer formulation or a genuine consequence of a nearly uniform $\hat{H}$ is the sharpest open question this sweep raises (Section 12).

9.3 The winter jet reaches four times the observed value

Under SB08 forcing on the positive-stability reference, as y_0 goes from 1400 to 2800 km:

- The winter (southern) jet strengthens from 91.6 to 128.8 m s^{-1} while the summer jet weakens from 26.9 to 12.3 m s^{-1} .
- Equatorial zonal wind becomes strongly easterly, from -25.4 to -54.2 m s^{-1} .

- Peak $|v|$ more than doubles, 4.38 to 9.50 m s^{-1} .
- The domain stops raining everywhere: the raining fraction falls from 1.000 to 0.845, so even a positive-stability state develops a dry zone once the forcing is far enough off the equator.

The directions are all as expected for a solstitial circulation, and the magnitudes are not. A 129 m s^{-1} winter jet is four times the observed zonal-mean maximum of 31 m s^{-1} , and -54 m s^{-1} of equatorial easterly is roughly three times what the solstice season shows. Section 5 takes the plausibility question up properly; the point here is that the off-equatorial overshoot compounds the jet overshoot the equinoctial runs already have.

9.4 On the negative-stability reference the rain leaves the tropics

The same forcing on the negative-stability reference ($a = 0.85$, $W_c = 50$) does something qualitatively different. At $y_0 = 700 \text{ km}$ the rain maximum is at 3.44 Mm, and by $y_0 = 2800 \text{ km}$ it is at 5.28 Mm, which is 46° equivalent latitude. The rain band is no longer near the forcing maximum at all; it has been relocated far poleward, and it now sits *poleward* of the energy flux equator (-0.01 to 2.48 Mm) rather than equatorward of it.

Every one of these eight runs fails the steadiness test of Section 11, so the positions above are maxima of a 1000-day mean field rather than of a settled state, and they should be read as such. The cleanest statement the block makes is that split itself: with the forcing off the equator, every run on the negative-stability reference keeps fluctuating and every run on the positive-stability reference settles.

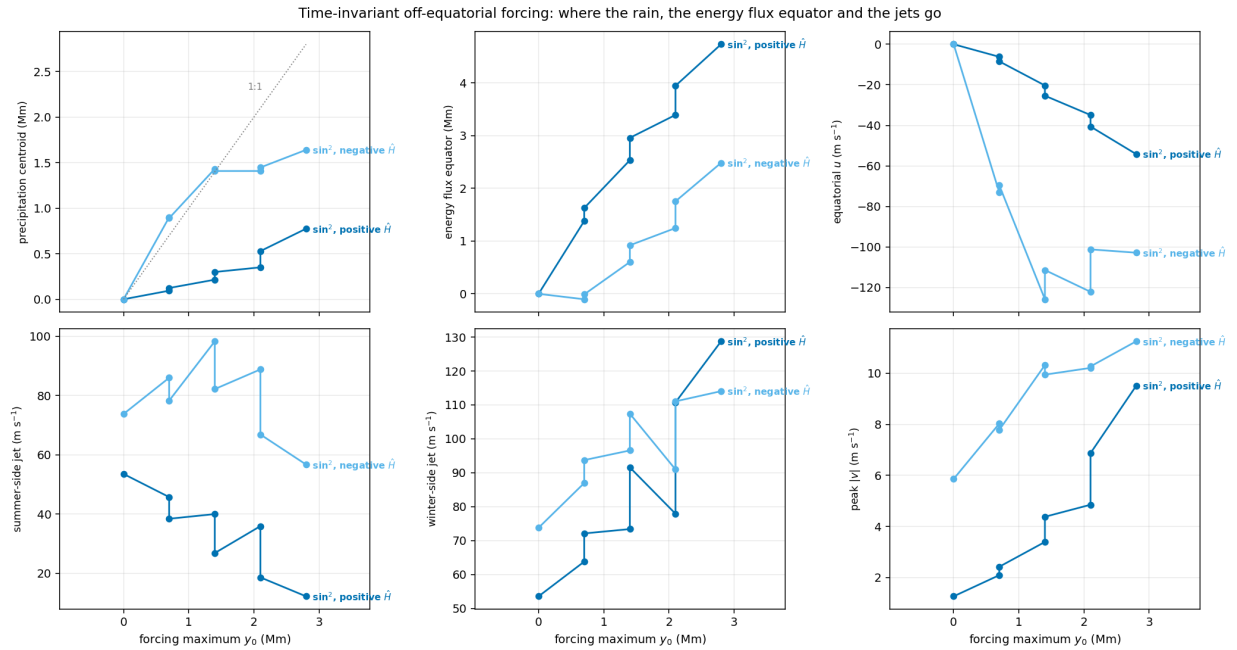


Figure 7: Time-invariant off-equatorial forcing. Rain position, energy flux equator, equatorial zonal wind, the two jets and the overturning strength against the forcing displacement y_0 , for both profile families and both reference states. The dotted 1:1 line in the first panel marks where the rain would sit if it followed the forcing exactly.

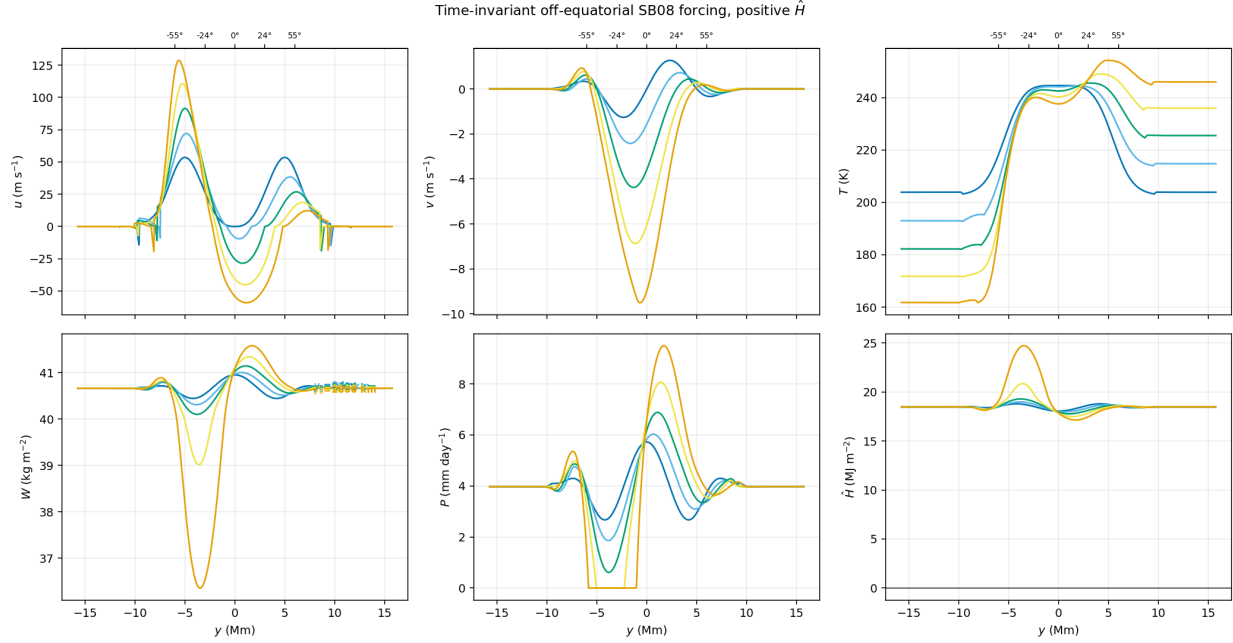


Figure 8: Equilibrium profiles for the SB08 off-equatorial series on the positive-stability reference, y_0 from 0 to 2800 km. The winter jet near $y = -5$ Mm strengthens to 129 m s^{-1} while the summer jet nearly disappears.

10 Forcing that migrates through a seasonal cycle

Nineteen runs vary y_0 in time with the SB08 profile: five amplitudes from 350 to 2800 km at a 360-day period, five periods from 90 to 1440 days at 1400 km amplitude, and the square and tanh waveforms, each on both reference states, plus one $n_y = 1601$ twin. Each is integrated for 2880 days plus at least six full cycles, and the last two complete cycles are compared as the repeatability check. All amplitudes and lags come from a least-squares fit at the forcing frequency, which is more robust on these series than a peak-finding estimate.

run	amp. (km)	period (d)	$\hat{H}$	centroid amp. (km)	gain	lag (d)	summer jet (m s ⁻¹)	winter jet (m s ⁻¹)
M_P_amp1400_p0090	1400	90	+	1192	0.852	3.1	70.2	70.2
M_P_amp1400_p0180	1400	180	+	1254	0.896	4.1	82.1	82.1
M_P_amp0350_p360	350	360	+	390	1.115	3.9	61.3	61.3
M_P_amp0700_p360	700	360	+	753	1.076	4.3	70.0	70.0
M_P_amp1400_p360	1400	360	+	1288	0.920	5.3	89.1	89.1
M_P_amp1400_p360_tanh	1400	360	+	1265	0.906	5.1	91.0	91.0
M_P_amp1400_p360_square	1400	360	+	1105	0.972	3.8	91.4	91.4
O_P_amp1400_p360_ny1601	1400	360	+	1287	0.920	5.3	89.1	89.1
M_P_amp2100_p360	2100	360	+	1626	0.774	6.0	108.3	108.3
M_P_amp2800_p360	2800	360	+	1957	0.699	6.2	126.7	126.7
M_P_amp1400_p0720	1400	720	+	1311	0.937	5.9	91.1	91.1
M_P_amp1400_p1440	1400	1440	+	1321	0.944	6.0	91.4	91.4
M_N_amp1400_p0090	1400	90	-	4116	2.941	4.8	122.6	140.2
M_N_amp0350_p360	350	360	-	384	1.097	-162.4	120.1	110.3
M_N_amp0700_p360	700	360	-	2759	3.942	42.5	133.7	137.3
M_N_amp1400_p360	1400	360	-	3132	2.237	14.9	166.3	170.2
M_N_amp2100_p360	2100	360	-	3634	1.731	14.2	196.6	193.0
M_N_amp2800_p360	2800	360	-	5661	2.022	1.0	200.1	217.1
M_N_amp1400_p1440	1400	1440	-	2630	1.879	135.3	176.9	178.5

10.1 The positive-stability cycles repeat exactly, and the negative ones never repeat

The cleanest result of the block is the split. On the positive-stability reference, consecutive cycles differ by 0.0 km in rain-band position for nine of the ten runs, and 0.4 km for the square-wave one. That is a fully converged periodic solution. On the negative-stability reference every run differs from its own previous cycle by 10,889 to 14,656 km, which is most of the domain. Those are chaotic, and their fitted amplitudes and lags describe one arbitrary realisation. They are in the table and marked, and no conclusion rests on them.

10.2 The response is quasi-static, and the obvious null model is wrong by a factor of ten

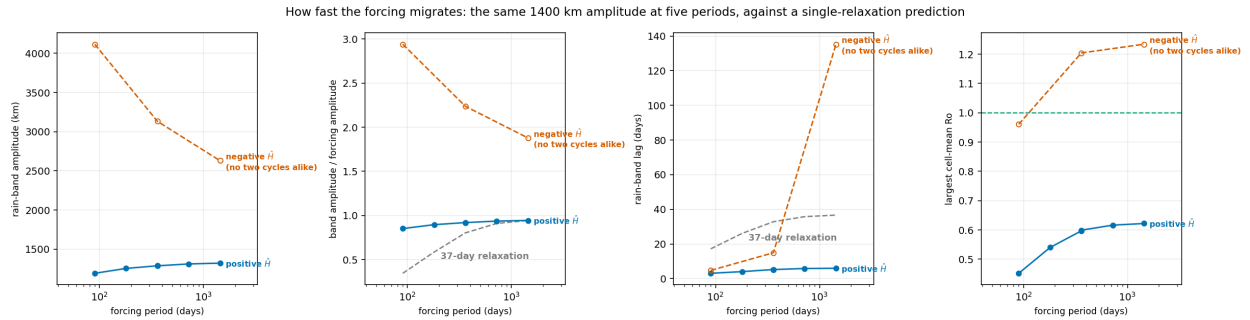


Figure 9: Response against forcing period at fixed 1400 km amplitude, with the prediction of a single relaxation on the model’s thermal timescale $\tau = 37$ d in grey. Dashed lines with open markers are the chaotic negative-stability runs, whose fitted values describe one realisation.

A single first-order relaxation on the model’s own thermal timescale $\tau = 37$ days is the natural null model. Driving $dX/dt = (F(t) - X)/\tau$ with $F = A \sin \omega t$ gives a response damped by

$1/\sqrt{1+\omega^2\tau^2}$ and lagged by $\arctan(\omega\tau)/\omega$. That predicts a lag of 17, 26, 33, 36 and 37 days at periods of 90, 180, 360, 720 and 1440 days, and a damping factor of 0.36, 0.61, 0.84, 0.95 and 0.99.

What the runs give at those five periods is a lag of 3.1, 4.1, 5.3, 5.9 and 6.0 days, and a gain of 0.852, 0.896, 0.920, 0.937 and 0.944. The lag is five to six times smaller than predicted at every period, and the damping at the shortest period is 0.85 where the null model says 0.36.

Physical reading (derived, then checked). The prediction fails because the rain band's position is not something the column temperature has to equilibrate to. It is set by where the forcing peaks, moderated by whatever pulls the ascent back toward the equator, and both act on the fast dynamical and moisture timescales rather than on τ . The check is that the seasonal gain should then equal the *steady* response at the same displacement. It does: the time-invariant SB08 run at $y_0 = 1400$ km puts the rain maximum at 1.082 Mm, a gain of 0.773, and the seasonal runs at 1400 km amplitude give a rain-maximum gain of 0.757 to 0.787 across the five periods, so the steady value sits inside that range. The migrating solution is tracking the sequence of steady states.

The same holds for the dynamics. Local Rossby number in the dominant cell, averaged over $|y|$ from 1 to 4 Mm, reaches 0.452, 0.540, 0.597, 0.616 and 0.622 at the five periods, and the steady $y_0 = 1400$ km run gives 0.624. At the two longest periods the migrating cycle matches the perpetual state to 1%, and at a 360-day period it falls 4% short. So there is no sign in this model of the seasonal cycle reaching an angular-momentum-conserving regime that the corresponding steady forcing does not, which is the mechanism Schneider and Bordoni identify for monsoon onset. If anything the migration keeps the cell slightly further from that limit.

10.3 What the amplitude does

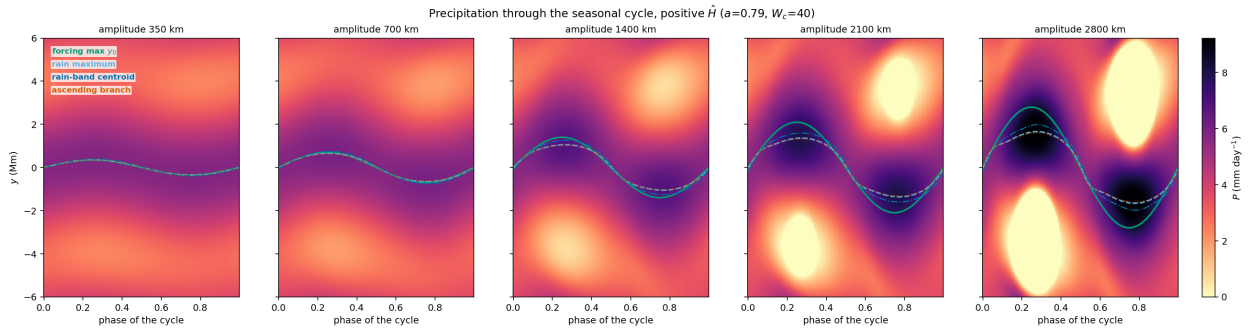


Figure 10: Precipitation through the composite cycle at five migration amplitudes on the positive-stability reference. Four position measures are drawn: the forcing maximum, the rain maximum, the centroid of the rain band, and the maximum of upper-level divergence. On the positive-stability reference they coincide closely.

Raising the migration amplitude from 350 to 2800 km at a 360-day period takes the summer-side jet from 61.3 to 126.7 m s^{-1} and the rain-band amplitude from 390 to 1957 km, so the gain falls from 1.115 to 0.699. The gain exceeding one at the smallest amplitudes is worth flagging as a small-signal result rather than interpreting: at 350 km amplitude the band moves 390 km, and the difference is of the order of ten gridpoints.

At amplitudes of 2100 and 2800 km a dry region opens in the winter hemisphere even on the positive-stability reference. The raining fraction dips to 0.905 and 0.844 at the solstices of the migrating runs, and the steady runs held at the same displacement give 0.905 and 0.845. So a far enough displacement produces a dry zone with no help from negative gross moist stability, and the migrating and steady cases produce the same one.

10.4 A migrating forcing against one held at the solstice

Four pairs compare the seasonal solution at its peak y_0 against the time-invariant run with the forcing held at that same y_0 :

	rain position, seasonal	steady	summer jet, seasonal	steady
700 km, positive $\hat{H}$	0.102 Mm	0.124 Mm	44.7 m s^{-1}	38.4 m s^{-1}
1400 km, positive $\hat{H}$	0.262	0.299	38.6	26.9
2100 km, positive $\hat{H}$	0.476	0.529	33.8	18.7

The migrating case reaches a rain band 22 to 53 km closer to the equator and a summer jet 6.3 to 15.1 m s^{-1} stronger than the state it would reach if the forcing stopped there. Both differences point the same way and have the same cause: the summer hemisphere has not had time to finish weakening its jet or to finish pulling the rain poleward. The peak precipitation is essentially identical between the two (6.02 against 6.03, 6.86 against 6.88, 8.02 against 8.06 mm day^{-1}), so the migration changes the geometry of the circulation without changing its rainfall intensity.

The fourth pair, at 1400 km on the negative-stability reference, differs far more (rain at 2.509 against 1.408 Mm, jet 113.8 against 82.2 m s^{-1} , peak precipitation 229.4 against 50.6 mm day^{-1}), and both members of that pair fail their respective steadiness and repeatability tests, so the comparison is between two fluctuating states and is not interpretable.

10.5 Waveform

Replacing the sinusoidal migration with a tanh or a square wave at 1400 km and 360 days changes very little: the gain is 0.920, 0.906 and 0.972 for sin, tanh and square, the lag 5.3, 5.1 and 3.8 days, and the summer jet 89.1, 91.0 and 91.4 m s^{-1} . The square wave’s smaller lag is the expected consequence of a forcing that spends most of the cycle stationary. The square-wave run is also the only positive-stability seasonal run whose consecutive cycles are not identical, and its 0.4 km difference is small enough to be the discrete jump itself rather than a failure to converge.

11 Timestep, grid spacing, and how far from equilibrium the runs are

11.1 The documented timestep is not enough everywhere

Before spending hours on long integrations I ran all 149 configurations for 100 days each and checked the output files. Twenty-two of them diverged, meaning the integration hit a NaN and stopped, in most cases within the first day or two. The repository’s note that $\Delta t = 30$ s is the V2 ceiling at $n_y = 801$ is right at the reference parameters and wrong as a general statement.

The 22 fall into two groups, and both make sense. One group has an unusually strong circulation: the three deeply negative-stability cases at $W_c = 60$ with $a \geq 0.83$ (r from 1.29 to 1.45), the two steepest radiative contrasts $\Delta_\theta^{\text{rad}} = 100$ and 110 K on the negative-stability reference, the two smallest static stabilities $\Delta_z = 30$ and 45 K, and the one case with the latent heating coefficient raised 25% above its physical value. The other group has a forcing maximum far off the equator: every run with $|y_0| \geq 1400$ km, and the $y_0 = 700$ km runs on the negative-stability reference.

Halving to $\Delta t = 15$ s rescued 16 of the 22 over a 300-day probe. The remaining six needed $\Delta t = 6$ s, and all five that stayed in the sweep ran clean at that value; the sixth was an $n_y = 1601$ off-equatorial case that would have cost 2.4 h on its own, so it was replaced (Section 9).

11.2 These are cold-start transients, not unstable equilibria

The distinction matters, because it says the equilibria themselves are reachable. A run whose forcing maximum sits at $y_0 = 1400$ km from the first timestep diverges at $\Delta t = 30$ s. A *seasonal* run whose forcing maximum walks out to the same 1400 km over 90 days does not: it survived the 100-day screen at $\Delta t = 30$ s. The model starts from $u = v = 0$ with θ at its target, and a large off-equatorial forcing imposed on that state produces a violent adjustment; the same forcing approached gradually does not.

Even so, every seasonal run in the sweep uses $\Delta t = 15$ s. Two reasons. One seasonal case, amplitude 2800 km on the negative-stability reference, diverged at day 800 rather than in the first days, so the screen would have passed it and the production run would have wasted its time. And holding Δt fixed across the seasonal block removes the timestep as a confounder when amplitudes and periods are compared against each other.

11.3 Timestep and grid spacing

Eight twin pairs test the numerics, each changing only Δt or n_y from its partner. The comparison is the pointwise difference after interpolating to the coarser grid, not a difference of summary numbers.

pair	$\max \Delta u $ (ms^{-1})	as % of peak u	where (Mm)	jet change
positive $\hat{H}$: Δt 30 to 15	0.0031	0.01%	−11.5	0.00%
negative $\hat{H}$: Δt 30 to 15	0.0087	0.01%	−7.4	0.00%
positive $\hat{H}$: n_y 801 to 1601	0.589	1.10%	−9.7	0.02%
negative $\hat{H}$: n_y 801 to 1601	7.20	9.76%	−7.4	0.04%
near-critical: n_y 801 to 1601	0.260	0.37%	−11.1	0.02%
off-equatorial: n_y 801 to 1601	29.98	41.58%	−7.5	0.02%
positive $\hat{H}$: n_y 801 to 401	7.58	14.14%	−9.6	0.01%
negative $\hat{H}$: n_y 801 to 401	49.86	67.58%	+7.4	0.30%

Two conclusions, and they point different ways.

The timestep is converged and the bulk quantities are converged in resolution.

Halving Δt changes u nowhere by more than 0.01% of its peak, which is negligible; that matters because 21 of the 151 runs use a smaller Δt than the rest for stability, and this says mixing timesteps across the sweep costs nothing. Across all eight pairs the jet moves by at most 0.30% and peak precipitation by at most 0.91%, so every scalar this report quotes is resolution-insensitive.

The profile in the westerly terminus region is not converged. Every one of the large differences sits at $|y|$ between 7 and 11 Mm, which is where the zonal wind turns over and the Heaviside gate on the eddy momentum flux switches. The 2026-07-12 $n_y = 1601$ probe recorded the same thing for the dry model, so this is a known property of the formulation rather than something the moisture introduces. What is new is the size on the negative-stability side: 9.8% between 801 and 1601, and 67.6% between 801 and 401. Any conclusion about the extratropical structure of an aggregated solution should be treated as resolution-dependent. Nothing in this report rests on one.

The off-equatorial pair is the weakest of the eight, and deliberately so: the $y_0 = 700$ km cold start diverges at $n_y = 1601$ unless $\Delta t = 6$ s, so both members of that pair run at 6 s and only for 2500 days, which is 2.2 drag timescales rather than 5. Both are equally unequilibrated, so their difference still measures grid sensitivity, and their bulk agreement is good (jet 38.44 against 38.45 m s^{-1} , peak $|v|$ 2.424 against 2.426).

11.4 How far from equilibrium the runs are

Every non-seasonal run is a fixed 5800-day integration, which is 5.01 drag timescales at $\epsilon_u = 10^{-8} \text{ s}^{-1}$, rather than a convergence-gated one. That choice makes the drift something this report measures instead of something a detector asserts.

For the 113 runs classified steady, the pointwise difference in u between the mean over days 3800–4800 and the mean over days 4800–5800 is between 7×10^{-10} and 3.9×10^{-3} of peak u , and the two runs at the top of that range are the 2500-day off-equatorial pair. Excluding those, the largest drift among the 111 full-length steady runs is 4.5×10^{-4} . The domain integral of precipitation equals that of evaporation to between 0.999 and 1.000 in the finite-volume measure for all 131, which is a constraint the integration has to satisfy at equilibrium and does. And no run had negative W at any timestep, which the MUSCL-limited transport is designed to prevent.

11.5 What the never-settling solutions look like

The 18 unsteady runs fall into two kinds. In 16 of them the jet magnitude itself swings, by 2.6 to 83.0 m s^{-1} over the trailing 2000 days. In two, both from the low- D end of the diffusivity ladder, the jet holds to under 0.7 m s^{-1} while the rain bands wander: their pointwise u difference between consecutive 1000-day means is 10% and 62%. A steadiness test on the jet alone would have passed those two, and their MSE budget residuals of 0.12 and 0.45 of the source are what gave them away.

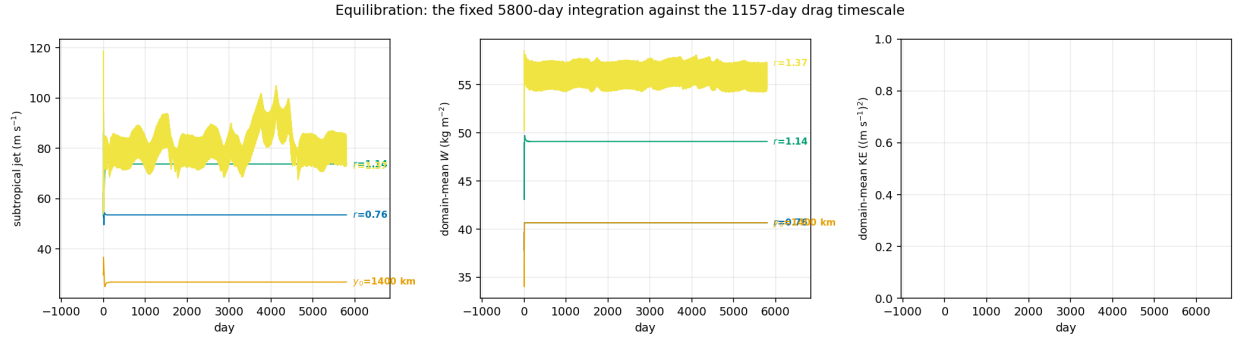


Figure 11: Equilibration of five runs spanning the transition, against the 1157-day drag timescale. The $r = 1.37$ run is one of the 18 that never settles; its jet and peak rain rate keep swinging for the whole integration.

The unsteadiness shows up in the profiles as large-amplitude oscillations in u poleward of about 5 Mm, visible on the two largest- W_c curves of Figure 3 reaching $\pm 80 \text{ m s}^{-1}$. Their wavelength is roughly 1.6 Mm, which is 40 gridpoints at $n_y = 801$, so they are not the $2\Delta y$ computational mode. Whether they are resolution-converged is untested: no $n_y = 1601$ twin was run above $r = 1.01$, and given the 9.8% difference the twins do show at $r = 1.145$ in exactly that region, they should not be trusted quantitatively.

12 Open questions, and the runs that would settle them

Each of these names what outcome would decide it, because a proposed run without that is not a test.

1. Why is the rain band 1.9 Mm from the energy flux equator, and is that a defect of the single layer? This is the sharpest question the sweep raises, because the standard framework

for ITCZ position ties the two together and here they come apart. The argument in Section 9 is that with $\hat{H}$ nearly uniform the total flux is essentially $v\hat{H}$, so it vanishes where v does, which is the cell boundary. That argument is testable: it predicts that the separation should shrink as $\hat{H}$ becomes less uniform. The negative-stability off-equatorial runs have strongly varying $\hat{H}$ and are all unsteady, so they cannot answer it. A run at intermediate r , around 1.01 to 1.05, with off-equatorial forcing, would have varying $\hat{H}$ and might still settle. If the separation shrinks with the variance of $\hat{H}$ the explanation holds; if it does not, something else is going on. This is the run I would do first.

2. Is the 40-gridpoint extratropical wave train real? The never-settling solutions oscillate at a wavelength of about 1.6 Mm poleward of 5 Mm. It is not the grid-scale computational mode, and the twins at $r = 1.145$ already show a 9.8% resolution sensitivity in exactly that region. An $n_y = 1601$ twin of one unsteady run, say $a = 0.85$, $W_c = 60$, would say whether the oscillation survives grid refinement. At $\Delta t = 15$ s and 5800 days that costs about an hour. If the wave train changes character with resolution, then the whole $r > 1.3$ corner of the map is describing the numerics rather than the physics.

3. Is there a second branch reachable from a drier start? The initial-condition block only tested moister starts, and found one attractor. Starting below W_c , so that nothing precipitates at $t = 0$, tests the other direction, and so does an asymmetric initial perturbation, which could find a single-cell state. Eight runs, about two hours. A negative result there would make the regime map genuinely a map of parameter space.

4. What is D , really? D has more leverage over the aggregated state than any other parameter, and the ERA5 calibration puts it only within a factor of 13, from 3×10^5 to 4×10^6 $\text{m}^2 \text{s}^{-1}$. Across that range the raining fraction of the negative-stability reference goes 0.54 to 0.72 and its peak rain rate 213 to 34 mm day^{-1} . No model result on that side of the boundary is better determined than D is. The ERA5 report's own recommendation, downloading the five empty ERA5 flux directories it lists, is the way to narrow it, and this sweep raises the value of doing so.

5. Can the jet overshoot be fixed without breaking the retarget? The jet is 1.5 to 2.6 times observed, and Section 5 traces most of it to the radiative retarget: 0.708 ms^{-1} per K of $\Delta_\theta^{\text{rad}}$, needing $\Delta_\theta^{\text{rad}} \approx 43$ K to reach the observed 31 ms^{-1} , which is below the radiative-convective 50 K and therefore not available. Two things are worth trying and neither was in this sweep. A larger v_d reduces the jet without touching the thermodynamics: at $v_d = 7.5 \text{ ms}^{-1}$ the jet is 47.5 ms^{-1} on the positive-stability reference, still 1.5 times observed, so a v_d ladder reaching 15 or 20 would say whether eddy momentum forcing alone can close the gap and at what cost to the overturning. And making the radiative background *colder* as well as steeper is the refinement SCIENCE.md §3.6 flags as not yet done; it is dynamically free as implemented, so it would not help the jet, which is itself worth stating so nobody tries it expecting otherwise.

6. Does the eddy energy transport gap have a fix inside this formulation? The model transports 0.38 MW m^{-1} of eddy energy at 24° against an observed 57.7, and the reason is structural: there is no eddy dry-static-energy flux in the equations, and the one eddy flux there is acts on a W field that local evaporation and precipitation hold nearly flat. Adding $\kappa_\theta \partial_y^2 \theta$ is already in the code and disabled by default. A ladder in κ_θ would show whether an eddy heat flux large

enough to matter for the energy budget leaves the circulation recognisable. That is a small change to test and would answer whether the gap is a missing term or a missing layer.

7. Which value of r should the model actually run at? The observations put the tropics near $r = 1$: the ERA5 estimate of W^* , 44 to 57 kg m⁻², straddles the model’s quiescent 50.66, and the observed band-mean $\hat{H}/\hat{S}$ of 0.26 to 0.32 is positive but small. This sweep says that the model’s behaviour on the two sides of $r = 1$ differs in kind, and that the boundary sits at $r \approx 0.94$ rather than 1. So the choice of a is not a detail. The ERA5-recommended $a \approx 0.79$ with $W_c = 40$ puts the model at $r = 0.76$, comfortably on the smooth side, and its jet is the least unrealistic of the configurations tested. Whether the real tropical atmosphere sits far enough onto the smooth side to justify that is a question about the observations, not about the model, and the largest uncertainty in it is the 34% floor on $\hat{S}$ from the barotropic mass adjustment that the ERA5 report documents.

A The model equations, as the code implements them

Transcribed from `ss09/sw_model.py` and `ss09/sw_config.py`, and cross-checked against `SCIENCE.md` §2–3. Not written from memory: the function that computes each term is named beside it, because an earlier report in this repository had four of these equations wrong.

The model is a single layer representing the zonal-mean upper troposphere on an equatorial β -plane, $y \in [-15751, +15751]$ km with rigid walls. The prognostic variables are the zonal wind u and potential temperature θ on n_y cell centres, the meridional wind v on the $n_y - 1$ interior faces (Arakawa C-grid), and the column water vapour W on the centres.

Zonal momentum (`StaggeredSWModel.du_dt`)

$$\frac{\partial u}{\partial t} = \underbrace{\beta y \bar{v}}_{\text{coriolis_term}} - \underbrace{\bar{v} \partial_y u|_{\text{upwind}}}_{\text{merid_advec_u}} - \underbrace{u (\partial_y v) \mathcal{H}(\partial_y v)}_{\text{vert_advec_u}} - \underbrace{\epsilon_u u}_{\text{rayleigh_drag_u}} - \underbrace{v_d \mathcal{H}(u) \text{sgn}(y) \partial_y u}_{\text{edd_mom_flux_div_u}}. \quad (5)$$

$\bar{v}$ is the face v averaged to the centres (`v_faces_to_centers`); $\partial_y v$ is the compact two-point face difference (`v_divergence_at_centers`). The eddy momentum flux divergence is *advective* in form, transporting u poleward at speed v_d wherever the flow is westerly, and its $\partial_y u$ uses a MUSCL reconstruction with monotonized-central limited slopes (`muscl_mc_du_dy`), which is the production stencil. Note the gate on the vertical momentum advection is the kinematic $\mathcal{H}(\partial_y v)$, SS09’s own form, not the thermodynamic $\mathcal{H}(\theta_E - \theta)$ used for dry and V1 runs: under latent heating a precipitating column runs *warmer* than its relaxation target, so the thermodynamic gate would switch off exactly where convection is strongest.

Meridional momentum (`StaggeredSWModel.dv_dt`), on the faces

$$2 \frac{\partial v}{\partial t} = -\beta y \bar{u} - \frac{gH}{T_0} \partial_y T + k_v \partial_y^2 v, \quad T = \theta (p_t/p_s)^{R/c_p} = \theta/1.6. \quad (6)$$

$\bar{u}$ is u averaged to the faces, $\partial_y T$ the compact face difference. The factor of 2 is the vertical integration with $v_{\text{top}} = -v_{\text{bottom}}$. The Laplacian uses a mirror wall ghost so that $v = 0$ at the walls.

Thermodynamics (`dtheta_dt`, `latent_heating`)

$$\frac{\partial \theta}{\partial t} = \frac{\theta_{\text{rad}} - \theta}{\tau} - \frac{d \Delta_z}{H} \partial_y v + \Lambda P, \quad \Lambda = \frac{L_v}{C}, \quad C = \frac{c_p \Pi \Delta p}{g}. \quad (7)$$

This equation is *not* gated. θ_{rad} is the same forcing-profile family as θ_E but at the steeper purely radiative contrast $\Delta_\theta^{\text{rad}}$ (`radiative_target_profile`), because the convective heating that V1's single relaxation bundled in is now explicit as ΛP and keeping both would double-count it.

Column water vapour (`moisture_transport_tendency`, `precipitation`)

$$\frac{\partial W}{\partial t} + \partial_y \left[-(2a - 1) v W - D \partial_y W \right] = E_0 - P, \quad P = \frac{(W - W_c)^+}{\tau_c}. \quad (8)$$

$-(2a - 1)v$ is the column transport velocity: the lower branch moves opposite to the upper-layer v , and with a fraction a of the column water vapour in the lower branch the column-integrated flux reduces to this one coefficient. The transport is finite-volume in flux form through the faces, with MUSCL-MC face values of W upwinded on the transport velocity's sign and zero flux at the walls, so transport moves no total water to roundoff. Advection is stepped at the central leapfrog level; D and P at the lagged $n - 1$ level.

Time stepping Leapfrog with an Asselin filter, coefficient 0.04. θ and W share the same Δt and filter coefficient, and the latent heating reads the *same* lagged P array the moisture sink uses, so $C\Lambda = L_v$ holds discretely and precipitation cancels exactly between the two budgets.

Column energetics Combining (7) and (8) with $\langle h \rangle = C\theta + L_v W$,

$$\frac{\partial \langle h \rangle}{\partial t} + \partial_y \left[v \hat{H} - L_v D \partial_y W \right] = \frac{C}{\tau} (\theta_{\text{rad}} - \theta) + L_v E_0, \quad \hat{H} = \hat{S} - L_v (2a - 1) W, \quad (9)$$

with the gross dry stability $\hat{S} = C d \Delta_z / H$. Precipitation is internal and does not appear.

Forcing profiles The $\sin^2$ profile (`Sin2Profile.__call__`) is

$$\theta_E = \theta_{00} - \Delta_y \sin^2 \left(\frac{\pi(y - y_0)}{2y_1} \right), \quad |y - y_0| < y_1, \quad (10)$$

a rigid translation of the equinoctial profile. The SB08 profile (`SB08Profile.__call__`) is

$$\theta_E = \theta_{00} - \Delta_y \left[\sin^2 \left(\frac{\pi y}{2y_1} \right) - 2 \sin \left(\frac{\pi y_0}{2y_1} \right) \sin \left(\frac{\pi y}{2y_1} \right) \right], \quad (11)$$

clamped to its $\pm y_1$ values outside. Its maximum sits at $y = y_0$, and its peak value *rises* above θ_{00} as y_0 moves off the equator, by $\Delta_y \sin^2(\pi y_0 / 2y_1)$: the SB08 family is not a rigid translation, and a solstitial SB08 forcing is stronger as well as displaced. Seasonal cycles vary y_0 in time and are available only for SB08. Under latent heating the profile these formulae describe is θ_{rad} and the contrast is $\Delta_\theta^{\text{rad}}$.

Parameter values Every run in the sweep uses $g = 9.81 \text{ m s}^{-2}$, $H = 16 \text{ km}$, $d = 4 \text{ km}$, $\beta = 2 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, $T_0 = 300 \text{ K}$, $\theta_{00} = 330 \text{ K}$, $y_1 = 9439 \text{ km}$, $\Delta_y = 50 \text{ K}$, $\tau = 37 \text{ d}$, $k_v = 7.786 \times 10^5 \text{ m}^2 \text{ s}^{-1}$, $\epsilon_u = 10^{-8} \text{ s}^{-1}$, $\kappa_\theta = 0$, and the production formulation (staggered v , EMFD gate on, MC stencil, kinematic vertical-advection gate, numba backend). The derived constants are $C = 5.1619 \times 10^6 \text{ J m}^{-2} \text{ K}^{-1}$, $\Lambda = 0.48431 \text{ K per kg m}^{-2} \text{ s}^{-1}$ and $\hat{S} = 7.7429 \times 10^7 \text{ J m}^{-2}$. Defaults for what the sweep varies: $a = 0.85$, $W_c = 50 \text{ kg m}^{-2}$, $\tau_c = 14400 \text{ s}$, $E_0 = 4.6 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ (3.97 mm day^{-1}), $D = 10^6 \text{ m}^2 \text{ s}^{-1}$, $v_d = 2.5 \text{ m s}^{-1}$, $\Delta_z = 60 \text{ K}$, $\Delta_\theta^{\text{rad}} = 75 \text{ K}$.

B Run inventory

The sweep is defined by `scripts/v2_sweep_manifest.py`, which is the single source of truth: it computes each run's parameters, records the timestep each configuration needs, and generates the command line. The table below is generated from it.

Table 1: Every run in the sweep.

run	block	n_y	$\Delta t \text{ (s)}$	what it varies
A.a075_wc30	A	801	30	GMS plane: a=0.75, W_c=30.0, r=0.495
A.a075_wc40	A	801	30	GMS plane: a=0.75, W_c=40.0, r=0.656
A.a075_wc50	A	801	30	GMS plane: a=0.75, W_c=50.0, r=0.818
A.a075_wc60	A	801	30	GMS plane: a=0.75, W_c=60.0, r=0.979
A.a079_wc30	A	801	30	GMS plane: a=0.79, W_c=30.0, r=0.574
A.a079_wc40	A	801	30	GMS plane: a=0.79, W_c=40.0, r=0.761
A.a079_wc50	A	801	30	GMS plane: a=0.79, W_c=50.0, r=0.949
A.a079_wc60	A	801	30	GMS plane: a=0.79, W_c=60.0, r=1.136
A.a083_wc30	A	801	30	GMS plane: a=0.83, W_c=30.0, r=0.653
A.a083_wc40	A	801	30	GMS plane: a=0.83, W_c=40.0, r=0.867
A.a083_wc50	A	801	30	GMS plane: a=0.83, W_c=50.0, r=1.080
A.a083_wc60	A	801	15	GMS plane: a=0.83, W_c=60.0, r=1.293
A.a085_wc30	A	801	30	GMS plane: a=0.85, W_c=30.0, r=0.693
A.a085_wc40	A	801	30	GMS plane: a=0.85, W_c=40.0, r=0.919
A.a085_wc50	A	801	30	GMS plane: a=0.85, W_c=50.0, r=1.145
A.a085_wc60	A	801	15	GMS plane: a=0.85, W_c=60.0, r=1.371
A.a087_wc30	A	801	30	GMS plane: a=0.87, W_c=30.0, r=0.733
A.a087_wc40	A	801	30	GMS plane: a=0.87, W_c=40.0, r=0.972
A.a087_wc50	A	801	30	GMS plane: a=0.87, W_c=50.0, r=1.210
A.a087_wc60	A	801	15	GMS plane: a=0.87, W_c=60.0, r=1.449
B.a085_wc35	B	801	30	fine W_c transect: W_c=35.0, r=0.806
B.a085_wc42	B	801	30	fine W_c transect: W_c=42.0, r=0.964
B.a085_wc44	B	801	30	fine W_c transect: W_c=44.0, r=1.009
B.a085_wc46	B	801	30	fine W_c transect: W_c=46.0, r=1.055
B.a085_wc55	B	801	30	fine W_c transect: W_c=55.0, r=1.258
C.r085_a075	C	801	30	matched r=0.85: a=0.75 needs W_c=51.99
C.r085_a079	C	801	30	matched r=0.85: a=0.79 needs W_c=44.73
C.r085_a083	C	801	30	matched r=0.85: a=0.83 needs W_c=39.23
C.r085_a087	C	801	30	matched r=0.85: a=0.87 needs W_c=34.91
C.r105_a075	C	801	30	matched r=1.05: a=0.75 needs W_c=64.38
C.r105_a079	C	801	30	matched r=1.05: a=0.79 needs W_c=55.41
C.r105_a083	C	801	30	matched r=1.05: a=0.83 needs W_c=48.61
C.r105_a087	C	801	30	matched r=1.05: a=0.87 needs W_c=43.28
D.P_dyr050	D	801	30	radiative contrast 50.0 K on the P reference
D.P_dyr060	D	801	30	radiative contrast 60.0 K on the P reference
D.P_dyr090	D	801	30	radiative contrast 90.0 K on the P reference
D.P_dyr100	D	801	30	radiative contrast 100.0 K on the P reference

run	block	n_y	Δt (s)	what it varies
D.P.dyr110	D	801	30	radiative contrast 110.0 K on the P reference
D.N.dyr050	D	801	30	radiative contrast 50.0 K on the N reference
D.N.dyr060	D	801	30	radiative contrast 60.0 K on the N reference
D.N.dyr090	D	801	30	radiative contrast 90.0 K on the N reference
D.N.dyr100	D	801	15	radiative contrast 100.0 K on the N reference
D.N.dyr110	D	801	15	radiative contrast 110.0 K on the N reference
E.a085_wc40_tc001800	E	801	30	$\tau_c=1800$ s at $a=0.85$, $W_c=40$, $r=0.906$
E.a085_wc40_tc007200	E	801	30	$\tau_c=7200$ s at $a=0.85$, $W_c=40$, $r=0.912$
E.a085_wc40_tc043200	E	801	30	$\tau_c=43200$ s at $a=0.85$, $W_c=40$, $r=0.949$
E.a085_wc40_tc086400	E	801	30	$\tau_c=86400$ s at $a=0.85$, $W_c=40$, $r=0.994$
E.a085_wc40_tc172800	E	801	30	$\tau_c=172800$ s at $a=0.85$, $W_c=40$, $r=1.084$
E.P.tc001800	E	801	30	$\tau_c=1800$ s on the P reference
E.P.tc172800	E	801	30	$\tau_c=172800$ s on the P reference
E.N.tc001800	E	801	30	$\tau_c=1800$ s on the N reference
E.N.tc172800	E	801	30	$\tau_c=172800$ s on the N reference
F.P.D0p00e00	F	801	30	$D=0$ m ² /s on the P reference
F.P.D2p50e05	F	801	30	$D=2.5e+05$ m ² /s on the P reference
F.P.D5p00e05	F	801	30	$D=5e+05$ m ² /s on the P reference
F.P.D2p00e06	F	801	30	$D=2e+06$ m ² /s on the P reference
F.P.D4p00e06	F	801	30	$D=4e+06$ m ² /s on the P reference
F.N.D0p00e00	F	801	30	$D=0$ m ² /s on the N reference
F.N.D2p50e05	F	801	30	$D=2.5e+05$ m ² /s on the N reference
F.N.D5p00e05	F	801	30	$D=5e+05$ m ² /s on the N reference
F.N.D2p00e06	F	801	30	$D=2e+06$ m ² /s on the N reference
F.N.D4p00e06	F	801	30	$D=4e+06$ m ² /s on the N reference
G.P.E2p30em05	G	801	30	$E_0=2.3e-05$ kg/m ² /s (1.99 mm/day) on P
G.P.E3p45em05	G	801	30	$E_0=3.45e-05$ kg/m ² /s (2.98 mm/day) on P
G.P.E6p90em05	G	801	30	$E_0=6.9e-05$ kg/m ² /s (5.96 mm/day) on P
G.P.E9p20em05	G	801	30	$E_0=9.2e-05$ kg/m ² /s (7.95 mm/day) on P
G.N.E2p30em05	G	801	30	$E_0=2.3e-05$ kg/m ² /s (1.99 mm/day) on N
G.N.E3p45em05	G	801	30	$E_0=3.45e-05$ kg/m ² /s (2.98 mm/day) on N
G.N.E6p90em05	G	801	30	$E_0=6.9e-05$ kg/m ² /s (5.96 mm/day) on N
G.N.E9p20em05	G	801	30	$E_0=9.2e-05$ kg/m ² /s (7.95 mm/day) on N
H.P.vd0p00	H	801	30	$v_d=0.0$ m/s on the P reference
H.P.vd1p25	H	801	30	$v_d=1.25$ m/s on the P reference
H.P.vd5p00	H	801	30	$v_d=5.0$ m/s on the P reference
H.P.vd7p50	H	801	30	$v_d=7.5$ m/s on the P reference
H.N.vd0p00	H	801	30	$v_d=0.0$ m/s on the N reference
H.N.vd1p25	H	801	30	$v_d=1.25$ m/s on the N reference
H.N.vd5p00	H	801	30	$v_d=5.0$ m/s on the N reference
H.N.vd7p50	H	801	30	$v_d=7.5$ m/s on the N reference
I.P.dz30	I	801	6	$\Delta z=30.0$ K on P, $r=1.523$
I.P.dz45	I	801	30	$\Delta z=45.0$ K on P, $r=1.015$
I.P.dz75	I	801	30	$\Delta z=75.0$ K on P, $r=0.609$
I.P.dz90	I	801	30	$\Delta z=90.0$ K on P, $r=0.508$
I.N.dz30	I	801	6	$\Delta z=30.0$ K on N, $r=2.290$
I.N.dz45	I	801	15	$\Delta z=45.0$ K on N, $r=1.527$
I.N.dz75	I	801	30	$\Delta z=75.0$ K on N, $r=0.916$
I.N.dz90	I	801	30	$\Delta z=90.0$ K on N, $r=0.763$
J.P.lam000	J	801	30	$\Lambda = 0.0 \times L_v/C$ on the P reference
J.P.lam025	J	801	30	$\Lambda = 0.25 \times L_v/C$ on the P reference
J.P.lam050	J	801	30	$\Lambda = 0.5 \times L_v/C$ on the P reference
J.P.lam075	J	801	30	$\Lambda = 0.75 \times L_v/C$ on the P reference
J.P.lam125	J	801	30	$\Lambda = 1.25 \times L_v/C$ on the P reference
J.N.lam000	J	801	30	$\Lambda = 0.0 \times L_v/C$ on the N reference
J.N.lam025	J	801	30	$\Lambda = 0.25 \times L_v/C$ on the N reference
J.N.lam050	J	801	30	$\Lambda = 0.5 \times L_v/C$ on the N reference

run	block	n_y	Δt (s)	what it varies
J_N_lam075	J	801	30	Lambda = 0.75 x L_v/C on the N reference
J_N_lam125	J	801	15	Lambda = 1.25 x L_v/C on the N reference
K_wc40_wi55	K	801	30	a=0.85, $W_c=40.0$, $W(t=0)=55.0$ (r=0.919)
K_wc40_wi70	K	801	30	a=0.85, $W_c=40.0$, $W(t=0)=70.0$ (r=0.919)
K_wc42_wi57	K	801	30	a=0.85, $W_c=42.0$, $W(t=0)=57.0$ (r=0.964)
K_wc42_wi72	K	801	30	a=0.85, $W_c=42.0$, $W(t=0)=72.0$ (r=0.964)
K_wc44_wi59	K	801	30	a=0.85, $W_c=44.0$, $W(t=0)=59.0$ (r=1.009)
K_wc44_wi74	K	801	30	a=0.85, $W_c=44.0$, $W(t=0)=74.0$ (r=1.009)
K_wc46_wi61	K	801	30	a=0.85, $W_c=46.0$, $W(t=0)=61.0$ (r=1.055)
K_wc46_wi76	K	801	30	a=0.85, $W_c=46.0$, $W(t=0)=76.0$ (r=1.055)
L_P_sin2.y0700	L	801	30	$\sin^2$ forcing peak at $y_0=700$ km on P
L_P_sin2.y1400	L	801	15	$\sin^2$ forcing peak at $y_0=1400$ km on P
L_P_sin2.y2100	L	801	15	$\sin^2$ forcing peak at $y_0=2100$ km on P
L_P_sb08.y0000	L	801	30	SB08 forcing peak at $y_0=0$ km on P
L_P_sb08.y0700	L	801	30	SB08 forcing peak at $y_0=700$ km on P
L_P_sb08.y1400	L	801	15	SB08 forcing peak at $y_0=1400$ km on P
L_P_sb08.y2100	L	801	15	SB08 forcing peak at $y_0=2100$ km on P
L_P_sb08.y2800	L	801	6	SB08 forcing peak at $y_0=2800$ km on P
L_N_sin2.y0700	L	801	15	$\sin^2$ forcing peak at $y_0=700$ km on N
L_N_sin2.y1400	L	801	15	$\sin^2$ forcing peak at $y_0=1400$ km on N
L_N_sin2.y2100	L	801	15	$\sin^2$ forcing peak at $y_0=2100$ km on N
L_N_sb08.y0000	L	801	30	SB08 forcing peak at $y_0=0$ km on N
L_N_sb08.y0700	L	801	15	SB08 forcing peak at $y_0=700$ km on N
L_N_sb08.y1400	L	801	15	SB08 forcing peak at $y_0=1400$ km on N
L_N_sb08.y2100	L	801	6	SB08 forcing peak at $y_0=2100$ km on N
L_N_sb08.y2800	L	801	6	SB08 forcing peak at $y_0=2800$ km on N
M_P_amp0350.p360	M	801	15	seasonal amplitude 350 km, 360-day period, P
M_P_amp0700.p360	M	801	15	seasonal amplitude 700 km, 360-day period, P
M_P_amp1400.p360	M	801	15	seasonal amplitude 1400 km, 360-day period, P
M_P_amp2100.p360	M	801	15	seasonal amplitude 2100 km, 360-day period, P
M_P_amp2800.p360	M	801	15	seasonal amplitude 2800 km, 360-day period, P
M_N_amp0350.p360	M	801	15	seasonal amplitude 350 km, 360-day period, N
M_N_amp0700.p360	M	801	15	seasonal amplitude 700 km, 360-day period, N
M_N_amp1400.p360	M	801	15	seasonal amplitude 1400 km, 360-day period, N
M_N_amp2100.p360	M	801	15	seasonal amplitude 2100 km, 360-day period, N
M_N_amp2800.p360	M	801	15	seasonal amplitude 2800 km, 360-day period, N
M_P_amp1400.p0090	M	801	15	seasonal period 90 days at amplitude 1400 km, P
M_P_amp1400.p0180	M	801	15	seasonal period 180 days at amplitude 1400 km, P
M_P_amp1400.p0720	M	801	15	seasonal period 720 days at amplitude 1400 km, P
M_P_amp1400.p1440	M	801	15	seasonal period 1440 days at amplitude 1400 km, P
M_N_amp1400.p0090	M	801	15	seasonal period 90 days at amplitude 1400 km, N
M_N_amp1400.p1440	M	801	15	seasonal period 1440 days at amplitude 1400 km, N
M_P_amp1400.p360.tanh	M	801	15	tanh seasonal waveform, amplitude 1400 km, 360-day period, P
M_P_amp1400.p360.square	M	801	15	square seasonal waveform, amplitude 1400 km, 360-day period, P
N_P_v1	N	801	30	moist V1 twin of the P reference: no latent heating, no retarget, thermodynamic gate
N_N_v1	N	801	30	moist V1 twin of the N reference: no latent heating, no retarget, thermodynamic gate
O_P_ny1601	O	1601	15	P reference at $ny=1601$, $dt=15$
O_P_ny401	O	401	60	P reference at $ny=401$, $dt=60$
O_P_dt15	O	801	15	P reference at $ny=801$, $dt=15$

run	block	n_y	Δt (s)	what it varies
O_N_ny1601	O	1601	15	N reference at ny=1601, dt=15
O_N_ny401	O	401	60	N reference at ny=401, dt=60
O_N_dt15	O	801	15	N reference at ny=801, dt=15
O_crit_ny1601	O	1601	15	near-critical a=0.85, W_c=44 at ny=1601, dt=15
O_P_sb08_y0700_ny1601	O	1601	15	off-equatorial SB08 y_0=700 km, P, at ny=1601, dt=15
O_P_amp1400_p360_ny1601	O	1601	15	seasonal amplitude 1400 km, P, at ny=1601, dt=15
O_P_sb08_y0700_ny0801_dt6	O	801	6	off-equatorial SB08 y_0=700 km, P, at ny=801, dt=6, 2500 d
O_P_sb08_y0700_ny1601_dt6	O	1601	6	off-equatorial SB08 y_0=700 km, P, at ny=1601, dt=6, 2500 d

C Commands

From the repository root, with the `claude-swm` conda environment:

```
# what the sweep is, and what it costs
python scripts/v2_sweep_manifest.py

# stability screen: every configuration for 100 days, 16 at a time
python scripts/v2_sweep_run.py --smoke 100 --nproc 16

# the sweep itself (about 2.5 h at 20 concurrent on berimbau)
python scripts/v2_sweep_run.py --nproc 20

# re-run only what failed or is missing
python scripts/v2_sweep_run.py --nproc 20 --skip-done

# diagnostics, every figure, and every number quoted above
python scripts/v2_sweep_figures.py
python scripts/v2_sweep_report_tables.py

# this document
latexmk -pdf -outdir=docs docs/moist_v2_sweep_report.tex
```

Output lives in `model_output/moist_v2_sweep/`, one directory per run holding `out.nc` and a final restart file, with `logs/` alongside and `status.jsonl` recording each run's exit state, wall time and stored days. The scalar table for every run is `all_runs.csv`, the quoted statistics are `numbers.json`, and the figures are in `figs/`.